



aws

SysOps Administrator

Associate Certification

300 Interview Questions & Answers

Comprehensive Exam Preparation Guide

Topics Covered:	10 Core AWS SysOps Domains
Total Questions:	300 Detailed Q&A Pairs
Difficulty:	Beginner to Advanced
Format:	A4 User-Friendly Content

Akhilesh Chaudhari

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300

Questions

10

Sections

All Levels

Difficulty

Exam Ready

Content

Section 1: Monitoring & Reporting (Q1-Q30)

Q 1**Interview Question**

What is Amazon CloudWatch?

Answer:

Amazon CloudWatch is AWS's monitoring and observability service. It collects metrics, logs, and events from AWS resources and applications in real time. You can set alarms, visualize dashboards, and automatically respond to changes in your infrastructure — making it the central hub for all monitoring tasks in AWS.

Q 2**Interview Question**

What are CloudWatch Metrics?

Answer:

CloudWatch Metrics are time-ordered data points sent by AWS services (like EC2, RDS, S3) to CloudWatch. Each metric has a namespace, name, and dimension. You can use these metrics to track performance, set threshold-based alarms, and make scaling decisions. Custom metrics can also be published from your own applications.

Q 3**Interview Question**

How do you create a CloudWatch Alarm?

Answer:

Go to CloudWatch > Alarms > Create Alarm. Select a metric, define a threshold condition (e.g., CPU > 80% for 5 minutes), choose the evaluation period, and configure the action — such as sending an SNS notification or triggering Auto Scaling. Alarms have three states: OK, ALARM, and INSUFFICIENT_DATA.

Q 4**Interview Question**

What is CloudWatch Logs?

Answer:

CloudWatch Logs is a service to store, monitor, and access log files from EC2 instances, Lambda functions, CloudTrail, and other AWS services. You install the CloudWatch Logs agent on EC2 to stream logs. Log groups organize logs, and log streams represent individual sources. Metric filters can extract data from logs.

Q 5

Interview Question

What is CloudWatch Log Insights?

Answer:

CloudWatch Log Insights is an interactive query tool that lets you analyze log data in CloudWatch Logs using a purpose-built query language. You can search and filter logs, aggregate data, visualize results as bar charts or time series, and save queries for reuse — all without exporting data.

Q 6

Interview Question

What is Amazon CloudTrail?

Answer:

AWS CloudTrail is a governance, compliance, and audit service that records API calls made in your AWS account. Every action — whether from the console, CLI, or SDK — is logged as an event with details like who made the request, when, from where, and what was changed. It is essential for security audits and troubleshooting.

Q 7

Interview Question

What is the difference between CloudWatch and CloudTrail?

Answer:

CloudWatch monitors the performance and health of AWS resources (CPU, memory, network metrics, logs). CloudTrail logs API activity — who did what in your account. Together they provide full observability: CloudWatch tells you what is happening to your resources, and CloudTrail tells you what actions caused it.

Q 8

Interview Question

What is AWS Config?

Answer:

AWS Config continuously records configuration changes to your AWS resources. It creates a timeline of resource configurations and lets you evaluate whether resources comply with your desired settings using Config Rules. It is ideal for auditing, compliance tracking, and understanding configuration drift over time.

Q 9**Interview Question**

What are CloudWatch Dashboards?

Answer:

CloudWatch Dashboards are customizable, reusable home pages in the CloudWatch console. You can add widgets showing metrics, alarms, logs, and text to build a single-pane view of your environment. Dashboards are global (cross-region) and can be shared with other AWS accounts.

Q 10**Interview Question**

How does CloudWatch handle high-resolution metrics?

Answer:

By default, CloudWatch metrics are published every 60 seconds (standard resolution). With high-resolution metrics, you can publish data every 1 second. This is useful for applications requiring near-real-time monitoring. High-resolution alarms can trigger as frequently as every 10 seconds, enabling faster auto-scaling responses.

Q 11**Interview Question**

What is CloudWatch Synthetics?

Answer:

CloudWatch Synthetics lets you create canaries — configurable scripts that monitor your endpoints and APIs even when there is no customer traffic. Canaries check availability, latency, and correctness of responses at scheduled intervals, helping you detect issues before your users do.

Q 12**Interview Question**

What is CloudWatch Container Insights?

Answer:

Container Insights collects, aggregates, and summarizes metrics and logs from containerized applications on Amazon ECS, EKS, and Kubernetes on EC2. It provides CPU, memory, disk, and network data at the cluster, node, pod, and task level — giving deep visibility into container performance.

Q 13**Interview Question**

What is AWS X-Ray?

Answer:

AWS X-Ray is a distributed tracing service that helps you analyze and debug production and distributed applications. It traces requests as they travel through your application components, identifies performance bottlenecks, and shows a service map of how components interact — crucial for microservices architectures.

Q 14**Interview Question**

What is the CloudWatch agent?

Answer:

The CloudWatch agent is a software package you install on EC2 instances (or on-premises servers) to collect system-level metrics and logs not available by default. It gathers memory usage, disk usage, swap, and custom application metrics, then sends them to CloudWatch for monitoring and alarming.

Q 15**Interview Question**

What are CloudWatch Anomaly Detection?

Answer:

CloudWatch Anomaly Detection uses machine learning to model the normal behavior of a metric over time. It automatically learns patterns like daily peaks or weekly cycles and creates dynamic alarm bands. You set alarms that trigger when a metric goes outside this expected range rather than a fixed threshold.

Q 16**Interview Question**

How do you monitor EC2 instance memory in CloudWatch?

Answer:

EC2 does not send memory metrics to CloudWatch by default because memory data resides inside the OS. To monitor memory, install the CloudWatch agent on the instance, configure it to collect memory metrics, and it will push custom metrics (mem_used_percent) to CloudWatch. You can then create alarms and dashboards on these metrics.

Q 17**Interview Question**

What is Amazon EventBridge?

Answer:

Amazon EventBridge (formerly CloudWatch Events) is a serverless event bus that connects AWS services, SaaS apps, and your custom applications. It routes events based on rules you define to targets like Lambda, SNS, SQS, or Step Functions — enabling event-driven architectures and automated operational responses.

Q 18**Interview Question**

What are CloudWatch Metric Filters?

Answer:

Metric Filters let you extract numerical values from log data in CloudWatch Logs. For example, you can count the occurrences of the word ERROR in your application logs and publish it as a CloudWatch metric. This turns unstructured log data into actionable metrics you can alarm on or chart.

Q 19**Interview Question**

What is AWS Personal Health Dashboard?

Answer:

AWS Personal Health Dashboard provides personalized alerts and remediation guidance when AWS events affect your resources. Unlike the Service Health Dashboard (which shows all AWS services globally), PHD shows only issues relevant to your account — such as scheduled maintenance or service degradations impacting your specific resources.

Q 20**Interview Question**

What is CloudWatch RUM?

Answer:

CloudWatch Real User Monitoring (RUM) collects performance data about real user interactions with your web application. It captures metrics like page load times, JavaScript errors, and HTTP errors from actual browser sessions, giving you insight into the end-user experience rather than just server-side performance.

Q 21**Interview Question****What is a CloudWatch Alarm composite alarm?****Answer:**

A Composite Alarm combines the states of multiple individual alarms using AND/OR logic. Instead of being notified about each individual alarm separately, you get a single notification when a specific combination of alarm states occurs — reducing alert fatigue and enabling more meaningful, context-aware alerting.

Q 22**Interview Question****How do you set up cross-account CloudWatch monitoring?****Answer:**

CloudWatch supports cross-account observability. You designate one account as the monitoring account and share data from source accounts using resource policies. The monitoring account can view metrics, logs, and traces from all linked accounts in a unified view — ideal for centralized operations in a multi-account AWS organization.

Q 23**Interview Question****What are CloudWatch Embedded Metric Format (EMF) logs?****Answer:**

EMF allows you to embed custom metrics inside structured log events using a specific JSON format. CloudWatch automatically extracts these embedded metrics and publishes them as CloudWatch Metrics asynchronously. This is particularly useful for Lambda functions where you can log and track business metrics in the same write operation.

Q 24**Interview Question****What is the CloudWatch retention policy for logs?****Answer:**

By default, CloudWatch Logs are stored indefinitely. You should configure a retention policy (from 1 day to 10 years) on each log group to control costs. Older logs can be exported to S3 for cheaper long-term archival. Configuring appropriate retention is a best practice for cost optimization and compliance.

Q 25**Interview Question**

How do you export CloudWatch Logs to S3?

Answer:

In the CloudWatch console, select the log group, choose Actions > Export data to S3. You specify the time range, destination S3 bucket, and optional prefix. The bucket must have a resource-based policy allowing CloudWatch to write to it. Export is a one-time operation — for continuous streaming, use a Kinesis Data Firehose subscription.

Q 26**Interview Question**

What is Amazon Managed Grafana?

Answer:

Amazon Managed Grafana is a fully managed service for Grafana — an open-source analytics platform. It integrates natively with AWS data sources like CloudWatch, Prometheus, X-Ray, and OpenSearch. SysOps administrators use it to build rich, interactive dashboards for operational monitoring without managing Grafana servers.

Q 27**Interview Question**

What metrics does CloudWatch collect from ELB?

Answer:

For Application Load Balancers, CloudWatch collects metrics like RequestCount, TargetResponseTime, HTTPCode_Target_5XX_Count, ActiveConnectionCount, and HealthyHostCount. For Classic Load Balancers, metrics include Latency, SurgeQueueLength, and SpilloverCount. These help you understand traffic patterns and identify backend health issues.

Q 28**Interview Question**

What is the GetMetricStatistics API?

Answer:

GetMetricStatistics is a CloudWatch API call that retrieves statistical data for a metric over a specified time period. You can request Average, Sum, Minimum, Maximum, or SampleCount statistics. It is useful for custom reporting and automation scripts that need historical metric data programmatically.

Q 29**Interview Question****How do you monitor AWS billing with CloudWatch?****Answer:**

Enable billing alerts in your AWS account settings (must be enabled in us-east-1). CloudWatch then receives billing metrics like EstimatedCharges. You can create billing alarms that notify you via SNS when your estimated charges exceed a defined threshold — an easy way to prevent unexpected cost overruns.

Q 30**Interview Question****What is CloudWatch ServiceLens?****Answer:**

CloudWatch ServiceLens unifies CloudWatch metrics and logs with AWS X-Ray traces to give you a complete, correlated view of your application health. The service map visualizes dependencies between components. You can drill from a high-level health indicator down to individual traces to understand the root cause of issues.

Section 2: High Availability & Elasticity (Q31-Q60)

Q 31**Interview Question**

What is Auto Scaling in AWS?

Answer:

AWS Auto Scaling automatically adjusts the number of EC2 instances (or other resources) in response to demand. You define minimum, desired, and maximum capacity. Scaling policies trigger based on CloudWatch alarms or predictive analytics — ensuring your application always has the right capacity without manual intervention.

Q 32**Interview Question**

What are the types of Auto Scaling policies?

Answer:

The main types are: Simple Scaling (add/remove instances based on a single alarm), Step Scaling (scale by different amounts based on alarm breach size), Target Tracking Scaling (maintain a target metric like 50% CPU), and Predictive Scaling (uses ML to forecast demand and pre-scales proactively).

Q 33**Interview Question**

What is a Launch Template vs Launch Configuration?

Answer:

Launch Templates are the modern replacement for Launch Configurations. Templates support versioning, multiple instance types, mixed On-Demand and Spot instances, and more EC2 features. Launch Configurations are legacy and do not support these advanced features. AWS recommends using Launch Templates for all new Auto Scaling groups.

Q 34**Interview Question**

What is an Elastic Load Balancer?

Answer:

An Elastic Load Balancer distributes incoming application traffic across multiple targets (EC2 instances, containers, Lambda functions) in one or more Availability Zones. It improves availability and fault tolerance. AWS offers three types: Application Load Balancer (HTTP/HTTPS), Network Load Balancer (TCP/UDP), and Gateway Load Balancer (for virtual appliances).

Q 35**Interview Question****What is the difference between ALB and NLB?****Answer:**

Application Load Balancer operates at Layer 7 (HTTP/HTTPS), supporting path-based and host-based routing, WebSockets, and HTTP/2. It is ideal for web applications. Network Load Balancer operates at Layer 4 (TCP/UDP/TLS), handling millions of requests per second with ultra-low latency — best for real-time applications requiring extreme performance.

Q 36**Interview Question****What are Availability Zones?****Answer:**

Availability Zones are distinct physical data center locations within an AWS Region, each with independent power, cooling, and networking. Deploying resources across multiple AZs protects against single-datacenter failures. Best practice is to deploy in at least two AZs for high availability. AZs within a region are connected via low-latency private networks.

Q 37**Interview Question****What is a Target Group in ALB?****Answer:**

A Target Group is a logical grouping of targets (EC2 instances, ECS tasks, Lambda functions, or IP addresses) that the load balancer routes requests to. Each target group has its own health check configuration. One ALB can route different paths or hostnames to different target groups — enabling microservices architectures.

Q 38**Interview Question****What is sticky sessions in ELB?**

Answer:

Sticky sessions (session affinity) configure the load balancer to route a user's requests to the same target for the duration of their session. ALB uses application-based cookies or load balancer-generated cookies. While useful for stateful apps, sticky sessions can create uneven load distribution. Stateless designs are preferred for scalability.

Q 39**Interview Question****What is Cross-Zone Load Balancing?****Answer:**

By default, a load balancer distributes traffic only to targets in the same AZ as the load balancer node. With cross-zone load balancing enabled, each node distributes traffic evenly across all registered targets in all AZs. This prevents uneven distribution when AZs have different numbers of instances. ALB has this enabled by default.

Q 40**Interview Question****What is connection draining / deregistration delay?****Answer:**

Connection draining (called Deregistration Delay in ALB/NLB) ensures that the load balancer stops sending new requests to a deregistering target but allows in-flight requests to complete. The default delay is 300 seconds (up to 3600). Setting a short delay (20-30s) is recommended for Lambda targets or fast-draining services.

Q 41**Interview Question****What is an Auto Scaling Lifecycle Hook?****Answer:**

Lifecycle Hooks pause an EC2 instance during Auto Scaling launch or termination to allow custom actions. During launch, you might install software or warm up a cache before the instance receives traffic. During termination, you might flush logs or drain connections. The instance stays in a wait state until you complete the action or the timeout occurs.

Q 42**Interview Question****What is a Scheduled Scaling Action?**

Answer:

Scheduled Scaling lets you scale your Auto Scaling group at a specific time. For example, scale out to 20 instances every weekday morning at 8 AM and scale in to 5 instances at 8 PM. This is ideal for predictable load patterns like business-hours traffic or batch jobs — allowing proactive capacity management.

Q 43**Interview Question****What is the health check grace period in Auto Scaling?****Answer:**

The health check grace period is the amount of time Auto Scaling waits after an instance launches before it starts checking the instance's health. This gives the instance time to initialize (install packages, start services). If health checks start too early, a starting instance may be incorrectly terminated. Default is 300 seconds.

Q 44**Interview Question****What is an Elastic IP address?****Answer:**

An Elastic IP address is a static, public IPv4 address that you can allocate to your AWS account and associate with an EC2 instance or NAT Gateway. Unlike dynamic public IPs that change on stop/start, an Elastic IP persists. It enables you to remap the address to another instance quickly in case of failure — supporting high-availability architectures.

Q 45**Interview Question****What is Warm Pools in Auto Scaling?****Answer:**

Warm Pools allow you to pre-initialize EC2 instances and keep them in a paused (stopped or running) state, ready to be added to the Auto Scaling group quickly during scale-out events. This significantly reduces launch latency for instances that have complex initialization steps — ideal for applications needing fast scale-out.

Q 46**Interview Question****What is an ELB health check?**

Answer:

Load balancer health checks periodically test the health of registered targets. For ALB, you configure a path (e.g., /health), port, protocol, success codes, and thresholds. If a target fails the configured number of checks, the load balancer stops sending traffic to it. Once it passes the healthy threshold, traffic resumes automatically.

Q 47**Interview Question****What is Route 53 health checks?****Answer:**

Route 53 health checks monitor the health of endpoints (IP addresses or domain names) and can automatically fail over DNS routing if an endpoint becomes unhealthy. They support HTTP, HTTPS, and TCP checks. Calculated health checks aggregate child health checks — useful for monitoring groups of resources in active-active or active-passive setups.

Q 48**Interview Question****What is a Multi-AZ deployment for RDS?****Answer:**

In Multi-AZ, Amazon RDS automatically provisions and maintains a synchronous standby replica in a different AZ. If the primary instance fails, RDS performs an automatic failover to the standby within 1-2 minutes. The failover is transparent to applications because the DNS endpoint stays the same. It is used for high availability, not for read scaling.

Q 49**Interview Question****What is an Auto Scaling Cooldown period?****Answer:**

A cooldown period is a configurable duration after a scaling activity during which Auto Scaling does not launch or terminate additional instances. This prevents Auto Scaling from rapidly adding or removing instances before the effects of a previous scaling activity take hold. Simple Scaling policies have a default cooldown of 300 seconds.

Q 50**Interview Question**

What is a Network ACL vs Security Group?

Answer:

Security Groups are stateful (return traffic is automatically allowed), operate at the instance level, and only support allow rules. Network ACLs are stateless (return traffic must be explicitly allowed), operate at the subnet level, and support both allow and deny rules. NACLs are evaluated in rule number order; Security Groups evaluate all rules.

Q 51**Interview Question**

What is AWS Global Accelerator?

Answer:

AWS Global Accelerator is a networking service that uses the AWS global network to route user traffic to the optimal AWS endpoint. It provides two static anycast IP addresses as fixed entry points. It improves availability and performance for global applications by routing traffic away from congested internet paths to AWS's backbone network.

Q 52**Interview Question**

What is an Application Load Balancer listener rule?

Answer:

Listener rules determine how the ALB routes requests from a client to one or more target groups. Rules are evaluated in priority order. You can route based on host headers (virtual hosting), URL paths, HTTP methods, query strings, source IP, and HTTP headers — enabling sophisticated traffic distribution and blue/green deployments.

Q 53**Interview Question**

What is EC2 Auto Recovery?

Answer:

EC2 Auto Recovery is a CloudWatch alarm action that automatically recovers an EC2 instance if it becomes impaired due to an underlying hardware failure. The recovered instance has the same instance ID, IP addresses, EBS volumes, and metadata. It only works with instances backed by EBS (not instance store) and on supported instance types.

Q 54**Interview Question**

What is a Gateway Load Balancer?

Answer:

Gateway Load Balancer helps you deploy, scale, and manage third-party virtual network appliances (firewalls, intrusion detection systems, deep packet inspection) in AWS. It uses the GENEVE protocol to forward traffic transparently to appliances and then sends the inspected traffic back to its original destination — ideal for security architectures.

Q 55**Interview Question**

What is Amazon EFS and how does it support HA?

Answer:

Amazon Elastic File System is a fully managed, scalable NFS file system. It stores data redundantly across multiple AZs within a region, so even if an AZ fails, data remains available. Multiple EC2 instances across different AZs can mount the same EFS file system simultaneously, making it ideal for shared storage in HA applications.

Q 56**Interview Question**

What is the difference between horizontal and vertical scaling?

Answer:

Horizontal scaling (scaling out/in) means adding or removing instances to distribute load — it is the AWS preferred approach since it improves availability and is nearly limitless. Vertical scaling (scaling up/down) means increasing the size of an existing instance (more CPU/RAM) — it has limits, requires downtime, and does not inherently improve availability.

Q 57**Interview Question**

What is Elastic Fabric Adapter (EFA)?

Answer:

EFA is a network interface for EC2 instances that enables high-performance computing (HPC) and machine learning applications. It provides OS-bypass capability, significantly reducing latency and increasing throughput for tightly coupled, distributed applications. EFA is ideal for MPI workloads and is available on specific instance types like c5n, p3dn, and Hpc instances.

Q 58**Interview Question****What is ALB Access Logs?****Answer:**

ALB Access Logs capture detailed information about requests sent to the load balancer — including client IP, latency, request path, and server response. Logs are delivered to S3 in compressed format. They are disabled by default. Enabling them is important for troubleshooting, security analysis, and understanding traffic patterns in production.

Q 59**Interview Question****What is an Auto Scaling instance refresh?****Answer:**

Instance Refresh is an Auto Scaling feature that lets you perform a rolling update of instances in your group when you update the launch template (e.g., new AMI). You specify the minimum healthy percentage and warm-up time. Auto Scaling gradually replaces old instances with new ones, ensuring your fleet runs the latest configuration with minimal downtime.

Q 60**Interview Question****What is Amazon ElastiCache and how does it support availability?****Answer:**

ElastiCache provides in-memory caching using Redis or Memcached. For Redis, you can enable Multi-AZ with automatic failover — the primary node is replicated to a read replica in another AZ, and if the primary fails, ElastiCache promotes the replica automatically. This reduces database load and provides sub-millisecond response times for cached data.

Section 3: Deployment & Provisioning (Q61-Q90)

Q 61**Interview Question**

What is AWS CloudFormation?

Answer:

AWS CloudFormation is an infrastructure-as-code service that lets you model, provision, and manage AWS resources using JSON or YAML templates. You describe your desired resources in a template, and CloudFormation handles creating, updating, and deleting them in the correct order. It enables repeatable, automated, and version-controlled infrastructure deployments.

Q 62**Interview Question**

What is an AMI (Amazon Machine Image)?

Answer:

An AMI is a pre-configured template that contains the OS, application code, libraries, and configuration needed to launch an EC2 instance. You can use AWS-provided AMIs, AWS Marketplace AMIs, or create custom AMIs by configuring an instance and creating an image from it. AMIs are region-specific but can be copied to other regions.

Q 63**Interview Question**

What is AWS Elastic Beanstalk?

Answer:

Elastic Beanstalk is a platform-as-a-service that automatically handles deployment, capacity provisioning, load balancing, auto-scaling, and health monitoring for your application. You simply upload your code, and Beanstalk deploys it on managed infrastructure. You retain full control over the underlying AWS resources but are freed from infrastructure management.

Q 64**Interview Question**

What is AWS OpsWorks?

Answer:

AWS OpsWorks is a configuration management service that uses Chef or Puppet to automate server configuration, deployment, and management. It offers three models: OpsWorks for Chef Automate, OpsWorks for Puppet Enterprise, and OpsWorks Stacks (custom solution). It is suitable for teams already using Chef/Puppet for configuration management.

Q 65**Interview Question**

What is EC2 User Data?

Answer:

EC2 User Data is a script that runs automatically when an EC2 instance launches for the first time. On Linux, it runs as root via cloud-init. It is commonly used to install packages, download code from S3, configure services, or bootstrap an application. User Data only runs once at launch unless explicitly configured to run on every boot.

Q 66**Interview Question**

What is EC2 Instance Metadata?

Answer:

Instance Metadata is data about your running EC2 instance that can be accessed from within the instance at the link-local address 169.254.169.254. You can retrieve the instance ID, AMI ID, public/private IP, IAM role credentials, availability zone, and more. It is essential for dynamic configuration and automation scripts running on EC2.

Q 67**Interview Question**

What is AWS Systems Manager (SSM)?

Answer:

AWS Systems Manager is a management service for operational tasks on AWS resources. Key features include Run Command (run scripts on EC2 without SSH), Session Manager (secure shell-less access), Patch Manager (automated patching), Parameter Store (secure configuration storage), and Automation (automated operational playbooks).

Q 68**Interview Question**

What is SSM Parameter Store?

Answer:

Parameter Store provides secure, hierarchical storage for configuration data and secrets like database passwords, API keys, and license codes. Parameters can be stored as String, StringList, or SecureString (encrypted with KMS). Applications retrieve parameters at runtime rather than hardcoding them, improving security and operational flexibility.

Q 69**Interview Question****What is AWS CodeDeploy?****Answer:**

AWS CodeDeploy automates code deployments to EC2 instances, on-premises servers, Lambda functions, and ECS. It supports in-place deployments (update existing instances) and blue/green deployments (replace existing instances). It integrates with CodePipeline for fully automated CI/CD pipelines and rolls back automatically on deployment failure.

Q 70**Interview Question****What is a blue/green deployment?****Answer:**

In a blue/green deployment, you maintain two identical environments: blue (current production) and green (new version). After testing the green environment, you switch traffic from blue to green — typically by updating a load balancer or DNS. If problems arise, you instantly roll back by redirecting traffic to blue. This eliminates downtime and reduces risk.

Q 71**Interview Question****What is a rolling deployment?****Answer:**

A rolling deployment updates instances in batches rather than all at once. A percentage of instances receive the new version while others continue serving the old version. This approach reduces deployment risk and maintains capacity during the update. If an issue is detected, the deployment stops and rollback restores remaining instances to the old version.

Q 72**Interview Question**

What is AWS Service Catalog?

Answer:

AWS Service Catalog allows organizations to create and manage catalogs of approved IT services for use on AWS. Administrators define portfolios of CloudFormation templates, and users can launch only pre-approved products — ensuring governance, compliance, and consistency while giving users self-service access to approved infrastructure.

Q 73**Interview Question**

What is EC2 Image Builder?

Answer:

EC2 Image Builder simplifies the creation, maintenance, validation, sharing, and deployment of customized, secure, and up-to-date Linux and Windows server images. You define image recipes (base image + components), and Image Builder automates building, testing, and distributing AMIs — replacing manual, error-prone AMI baking processes.

Q 74**Interview Question**

What is AWS Backup?

Answer:

AWS Backup is a fully managed service that centralizes and automates backups across AWS services including EC2, EBS, RDS, DynamoDB, EFS, FSx, and more. You create backup plans with schedules and retention policies. It provides a central console to manage backups, restore data, and ensure compliance with backup requirements across all services.

Q 75**Interview Question**

What is the difference between snapshot and AMI?

Answer:

An EBS Snapshot is a point-in-time backup of an EBS volume stored in S3. An AMI includes one or more snapshots of the EBS volumes plus launch permissions and a block device mapping. You use snapshots to back up and restore volumes; you use AMIs to launch new EC2 instances with a specific pre-configured OS and application environment.

Q 76**Interview Question**

What is an EC2 Placement Group?

Answer:

A Placement Group controls how EC2 instances are physically placed on hardware. Cluster placement groups pack instances closely in one AZ for low-latency HPC workloads. Spread placement groups distribute instances across distinct hardware to reduce correlated failures. Partition placement groups spread instances across logical partitions for large distributed systems like Hadoop.

Q 77**Interview Question**

What is AWS CloudFormation StackSets?

Answer:

CloudFormation StackSets extends stacks to deploy resources across multiple AWS accounts and regions with a single template. Administrators create StackSets from a management account, and CloudFormation deploys them to specified target accounts. This is ideal for enforcing consistent baseline infrastructure (security groups, IAM roles, logging) across an entire organization.

Q 78**Interview Question**

What is the EC2 Instance Connect?

Answer:

EC2 Instance Connect provides a simple and secure way to connect to EC2 instances using SSH without managing SSH keys. It pushes a one-time-use SSH public key to the instance's `authorized_keys` via the EC2 Instance Metadata Service. Access is controlled via IAM policies, and all connections are logged in CloudTrail.

Q 79**Interview Question**

What is an AWS CloudFormation Drift Detection?

Answer:

Drift Detection in CloudFormation compares the current configuration of stack resources against the expected configuration defined in the template. If someone manually modifies a resource outside of CloudFormation, it shows as drifted. You can then decide to import the change, update the template, or bring the resource back into compliance.

Q 80**Interview Question****What is AWS Elastic Container Registry (ECR)?****Answer:**

Amazon ECR is a fully managed Docker container registry that stores, manages, and deploys container images. It integrates with ECS, EKS, and Lambda. ECR supports image lifecycle policies to clean up old images, image scanning for vulnerabilities, cross-region and cross-account replication, and immutable image tags for deployment consistency.

Q 81**Interview Question****What is AWS CodePipeline?****Answer:**

AWS CodePipeline is a fully managed continuous delivery service that automates the build, test, and deploy phases of your release process. It connects source (CodeCommit, GitHub), build (CodeBuild), and deploy (CodeDeploy, Elastic Beanstalk, ECS) stages into a pipeline that triggers on every code change — enabling rapid, reliable software delivery.

Q 82**Interview Question****What is an EC2 Dedicated Host?****Answer:**

An EC2 Dedicated Host is a physical server fully dedicated to your use. It gives you visibility into the physical sockets and cores, which is required for certain software licenses (like Windows Server or Oracle) tied to physical hardware. You can use Dedicated Hosts for Bring-Your-Own-License (BYOL) scenarios to remain compliant.

Q 83**Interview Question****What are EC2 Spot Instances?****Answer:**

Spot Instances allow you to use spare AWS EC2 capacity at up to 90% discount compared to On-Demand prices. AWS can reclaim Spot Instances with a 2-minute warning when capacity is needed. They are ideal for fault-tolerant, flexible workloads like batch processing, data analysis, and CI/CD — but should not be used for critical, stateful workloads.

Q 84**Interview Question****What is AWS Proton?****Answer:**

AWS Proton is a managed service for deploying containerized and serverless applications. Platform teams define infrastructure templates for different environments and service types. Development teams use these templates to deploy their services without deep infrastructure knowledge — combining self-service agility with platform team governance.

Q 85**Interview Question****What is an EC2 Dedicated Instance?****Answer:**

A Dedicated Instance runs on hardware dedicated to a single customer (your account) but may share hardware with other non-Dedicated Instances in your account. Unlike Dedicated Hosts, you do not get visibility into the underlying hardware. Dedicated Instances ensure physical isolation from other customers — required for certain compliance regulations.

Q 86**Interview Question****What is AWS AppConfig?****Answer:**

AWS AppConfig (part of Systems Manager) enables you to deploy application configuration changes separately from code deployments. You define configurations, validators (Lambda or JSON Schema), deployment strategies, and environments. AppConfig validates configurations before deployment and monitors for errors — enabling safe, controlled configuration rollouts.

Q 87**Interview Question****What is AWS CloudFormation Change Sets?****Answer:**

Change Sets let you preview how proposed changes to a CloudFormation stack will affect existing resources before executing them. CloudFormation shows which resources will be added, modified, or deleted. This is critical for production stacks where accidental resource replacement (like deleting an RDS database) could cause data loss or downtime.

Q 88**Interview Question**

What is AWS License Manager?

Answer:

AWS License Manager helps you manage software licenses from vendors like Microsoft, SAP, Oracle, and IBM in AWS and on-premises environments. You create license configurations that define rules based on vCPUs, cores, or sockets. License Manager tracks usage and prevents non-compliant deployments — avoiding costly licensing violations and audits.

Q 89**Interview Question**

What is EC2 Hibernate?

Answer:

EC2 Hibernate saves the contents of RAM (in-memory state) to the root EBS volume and stops the instance. When you restart, the OS resumes exactly where it left off — all processes, data in RAM, and TCP connections remain intact. This enables much faster startup compared to a regular start and is useful for long-running processes or development environments.

Q 90**Interview Question**

What is AWS Fault Injection Simulator?

Answer:

AWS FIS is a managed chaos engineering service that helps you improve resilience by injecting controlled faults into your workloads. You can simulate CPU stress, memory pressure, network disruptions, and AZ failures. FIS integrates with CloudWatch to monitor impact and automatically halts experiments if predefined safety conditions are breached.

Section 4: Storage & Data Management (Q91-Q120)

Q 91**Interview Question**

What are the different S3 storage classes?

Answer:

S3 offers several storage classes: S3 Standard (frequently accessed data, 99.99% availability), S3 Standard-IA (infrequently accessed, lower cost), S3 One Zone-IA (single AZ, lower cost), S3 Glacier Instant Retrieval (archive with millisecond access), S3 Glacier Flexible Retrieval (hours retrieval), S3 Glacier Deep Archive (lowest cost, 12-hour retrieval), and S3 Intelligent-Tiering (auto-moves objects between tiers).

Q 92**Interview Question**

What is S3 Versioning?

Answer:

S3 Versioning keeps multiple versions of an object in a bucket. When enabled, every upload creates a new version, and deletes only add a delete marker rather than permanently removing the object. This protects against accidental deletion and overwrites. You can restore previous versions at any time. Versioning is required for S3 Replication.

Q 93**Interview Question**

What is S3 Cross-Region Replication (CRR)?

Answer:

CRR automatically replicates objects from a source bucket in one region to a destination bucket in a different region. It requires versioning enabled on both buckets. Uses include compliance (geographically separate copies), latency optimization (replicate closer to users), and disaster recovery. New objects are replicated; existing objects require S3 Batch Replication.

Q 94**Interview Question**

What is S3 Lifecycle Policy?

Answer:

S3 Lifecycle policies automate transitioning objects between storage classes or deleting them after a specified time. For example, move objects to Standard-IA after 30 days, to Glacier after 90 days, and delete after 365 days. This significantly reduces storage costs for data that becomes less frequently accessed over time.

Q 95**Interview Question**

What is S3 Object Lock?

Answer:

S3 Object Lock prevents objects from being deleted or overwritten for a fixed time (retention period) or indefinitely. It uses WORM (Write Once Read Many) protection. There are two modes: Governance (users with permission can override) and Compliance (no one, including root, can delete or modify until the retention period expires). Used for regulatory compliance.

Q 96**Interview Question**

What is Amazon EBS?

Answer:

Amazon Elastic Block Store provides persistent, high-performance block storage volumes for use with EC2 instances. Unlike instance store (which is ephemeral), EBS volumes persist independently from the instance lifecycle. EBS volumes are in a specific AZ and can only be attached to instances in the same AZ. They support snapshots for backup and replication.

Q 97**Interview Question**

What are the EBS volume types?

Answer:

EBS has several types: gp3 (General Purpose SSD, up to 16,000 IOPS, cost-effective), gp2 (older GP SSD), io2/io2 Block Express (Provisioned IOPS SSD for I/O-intensive workloads, up to 256,000 IOPS), st1 (Throughput Optimized HDD for big data streaming), and sc1 (Cold HDD for infrequently accessed data at lowest cost).

Q 98**Interview Question**

What is Amazon EFS?

Answer:

Amazon Elastic File System is a managed NFS file system for use with AWS cloud services and on-premises resources. It automatically grows and shrinks as files are added and removed. Multiple EC2 instances can mount EFS simultaneously. It supports two performance modes (General Purpose, Max I/O) and two storage classes (Standard, Infrequent Access).

Q 99**Interview Question****What is Amazon S3 Transfer Acceleration?****Answer:**

S3 Transfer Acceleration uses AWS CloudFront edge locations to speed up uploads to S3. Data uploaded to an edge location is transferred over the optimized AWS backbone network to the S3 bucket. It is beneficial for users uploading large files from distant geographic locations. You access it via a different endpoint (bucket-name.s3-accelerate.amazonaws.com).

Q 100**Interview Question****What is Amazon FSx?****Answer:**

Amazon FSx provides fully managed file systems optimized for specific use cases: FSx for Windows File Server (SMB protocol, supports Active Directory), FSx for Lustre (high-performance for HPC, ML, and media workflows, integrates with S3), FSx for NetApp ONTAP (enterprise NAS features), and FSx for OpenZFS (Linux-compatible, low-latency workloads).

Q 101**Interview Question****What is S3 Multipart Upload?****Answer:**

Multipart Upload allows you to upload large objects (recommended for files over 100 MB) in parts that are uploaded independently and in parallel. Once all parts are uploaded, S3 assembles them into the final object. Benefits include improved throughput, quick recovery from network issues (retry only failed parts), and the ability to start uploading before you know the total object size.

Q 102**Interview Question****What is S3 Select?****Answer:**

S3 Select lets you retrieve only the data you need from an object using SQL-like expressions. Instead of downloading an entire CSV, JSON, or Parquet file and filtering it in your application, S3 Select filters the data server-side and returns only the matching rows or columns. This reduces data transfer costs and improves application performance.

Q 103**Interview Question****What is EBS encryption?****Answer:**

EBS encryption uses AWS KMS keys to encrypt the data at rest on EBS volumes, snapshots, and all data moving between an EC2 instance and the volume. You can enable default encryption at the account level so all new EBS volumes are automatically encrypted. There is no performance impact for EC2 instances that support NVMe SSD volumes.

Q 104**Interview Question****What is an EBS Snapshot?****Answer:**

An EBS Snapshot is an incremental backup of an EBS volume stored durably in Amazon S3. Only blocks that have changed since the last snapshot are stored. Snapshots can be used to create new EBS volumes, copy data across regions, or share volumes with other AWS accounts. First snapshots copy all data; subsequent ones are incremental.

Q 105**Interview Question****What is Amazon S3 Intelligent-Tiering?****Answer:**

S3 Intelligent-Tiering automatically moves objects between four access tiers based on changing access patterns: Frequent Access, Infrequent Access (30+ days no access), Archive Instant Access (90+ days), and Archive Access (configurable). There is no retrieval fee. It is ideal for data with unknown or unpredictable access patterns, eliminating the guesswork in storage class selection.

Q 106**Interview Question**

What is AWS Storage Gateway?

Answer:

AWS Storage Gateway bridges on-premises environments with AWS cloud storage. It has three types: File Gateway (NFS/SMB interface to S3), Volume Gateway (iSCSI block storage backed by S3, with cloud snapshots), and Tape Gateway (virtual tape library backed by Glacier). It is used for backup, archiving, disaster recovery, and cloud bursting.

Q 107**Interview Question**

What is S3 Event Notifications?

Answer:

S3 Event Notifications can trigger automated actions when specific events occur in your bucket (object created, deleted, or restored). Notifications can be sent to Amazon SQS queues, SNS topics, or Lambda functions. Common uses include image processing pipelines, log analysis, backup workflows, and triggering downstream applications on new data arrival.

Q 108**Interview Question**

What is Amazon Data Lifecycle Manager (DLM)?

Answer:

Amazon DLM automates the creation, retention, and deletion of EBS snapshots and EBS-backed AMIs. You create lifecycle policies that define schedules (e.g., daily snapshot at 2 AM, retain 7 days) and apply tags to target specific volumes or instances. This replaces manual snapshot management and ensures consistent backup compliance.

Q 109**Interview Question**

What is S3 Requester Pays?

Answer:

With Requester Pays enabled on an S3 bucket, the requester (the person or application downloading data) pays the data transfer and request costs — rather than the bucket owner. The bucket owner still pays for storage. This is useful for public datasets where you want to share data without bearing the cost of others downloading it.

Q 110**Interview Question****What is Amazon EBS io2 Block Express?****Answer:**

io2 Block Express is the highest-performance EBS volume type, designed for mission-critical workloads requiring sub-millisecond latency. It supports up to 256,000 IOPS and 4,000 MB/s throughput, with up to 64 TiB volume size. It uses NVMe SSD and is built on the Nitro System — ideal for Oracle, SAP HANA, and other latency-sensitive databases.

Q 111**Interview Question****What is S3 pre-signed URLs?****Answer:**

Pre-signed URLs grant time-limited access to a private S3 object without requiring the requester to have AWS credentials. You generate a URL with your credentials, specifying the object and expiry time (up to 7 days for SDK, 12 hours for console). Anyone with the URL can access the object during the validity period — useful for secure file sharing.

Q 112**Interview Question****What is Amazon S3 Access Points?****Answer:**

S3 Access Points simplify managing data access for shared datasets. Instead of one complex bucket policy, each application or team gets its own access point with its own policy controlling which prefixes it can access and with what permissions. Access points can be VPC-only, restricting access to private networks only — improving security and governance.

Q 113**Interview Question**

What is the S3 Consistency Model?

Answer:

Since December 2020, Amazon S3 provides strong read-after-write consistency for all operations. After a successful write (PUT, DELETE, or overwrite), any subsequent read immediately returns the new version. This eliminates the need for workarounds in applications that previously had to handle eventual consistency — simplifying application development significantly.

Q 114**Interview Question**

What is AWS DataSync?

Answer:

AWS DataSync is a managed data transfer service for moving large amounts of data between on-premises storage, S3, EFS, FSx, and other AWS storage services. It handles data validation, encryption, compression, and network optimization automatically. DataSync is up to 10x faster than open-source tools and can transfer data over Direct Connect or the internet.

Q 115**Interview Question**

What is S3 Block Public Access?

Answer:

S3 Block Public Access provides four settings at the account or bucket level to prevent public access to S3 data: block new public ACLs, block public bucket policies, ignore existing public ACLs, and restrict public access from policies. Best practice is to enable all four settings at the account level unless you specifically need public access.

Q 116**Interview Question**

What is an EBS snapshot archive?

Answer:

EBS Snapshot Archive moves snapshots to a low-cost archive tier at up to 75% lower cost than standard snapshot storage. Archived snapshots must remain archived for a minimum of 90 days. Restoring an archived snapshot takes 24-72 hours. It is suitable for long-term retention of compliance snapshots that are rarely if ever needed for restore.

Q 117**Interview Question****What is Amazon S3 Batch Operations?****Answer:**

S3 Batch Operations lets you perform bulk operations on billions of S3 objects with a single request. Operations include copying objects, invoking Lambda for each object, restoring Glacier objects, replacing object tags, changing ACLs, and replicating existing objects. You provide an S3 inventory manifest as input and track job progress in the console.

Q 118**Interview Question****What is Amazon Glacier Vault Lock?****Answer:**

Glacier Vault Lock enforces compliance controls on an S3 Glacier vault using a lockable vault access policy. Once locked, the policy cannot be changed or deleted. It supports WORM policies (objects cannot be deleted before a retention period) and is used for regulatory requirements like SEC Rule 17a-4 that require immutable data storage.

Q 119**Interview Question****What is Amazon S3 Storage Lens?****Answer:**

S3 Storage Lens provides organization-wide visibility into object storage usage and activity. It delivers metrics about storage size, request count, cost optimization opportunities, and data protection status across all buckets and accounts in an AWS Organization. It highlights anomalies and helps you identify cost savings and security posture improvements.

Q 120**Interview Question****What is Amazon EBS Multi-Attach?****Answer:**

EBS Multi-Attach allows a single io1 or io2 EBS volume to be attached to up to 16 EC2 instances in the same AZ simultaneously. Each instance has full read and write permissions. This is useful for high-availability clustering applications (like Oracle RAC) that require concurrent write access to shared storage. Applications must manage concurrent writes carefully.

Section 5: Security & Compliance (Q121-Q150)

Q 121**Interview Question**

What is AWS IAM?

Answer:

AWS Identity and Access Management lets you control who (authentication) can do what (authorization) in your AWS account. IAM manages users, groups, roles, and policies. Policies are JSON documents that define permissions. Best practices include least privilege access, using roles for EC2/Lambda instead of access keys, enabling MFA, and rotating credentials regularly.

Q 122**Interview Question**

What is the difference between IAM Roles and IAM Users?

Answer:

An IAM User is a permanent identity with long-term credentials (password, access keys) associated with a specific person or application. An IAM Role is a temporary identity that can be assumed by AWS services (EC2, Lambda), applications, or users from other accounts. Roles provide temporary credentials — they are more secure than long-term access keys for workloads.

Q 123**Interview Question**

What is AWS KMS?

Answer:

AWS Key Management Service is a managed service for creating and controlling cryptographic keys. KMS keys are used to encrypt S3 objects, EBS volumes, RDS databases, and many other AWS services. Keys never leave KMS unencrypted. KMS integrates with CloudTrail to log every key use. You can create customer-managed keys (CMKs) for additional control.

Q 124**Interview Question**

What is an IAM Policy?

Answer:

An IAM policy is a JSON document that defines permissions. Policies specify which actions are allowed or denied on which resources under what conditions. There are identity-based policies (attached to users/groups/roles), resource-based policies (attached to S3 buckets, KMS keys), permission boundaries, and service control policies (SCPs) in AWS Organizations.

Q 125**Interview Question**

What is the AWS Shared Responsibility Model?

Answer:

AWS is responsible for security OF the cloud (physical infrastructure, hypervisor, networking hardware). You (the customer) are responsible for security IN the cloud: what you put in AWS, your operating systems, network configurations, IAM permissions, encryption of data, and application security. Understanding this division is fundamental for designing secure AWS architectures.

Q 126**Interview Question**

What is AWS WAF?

Answer:

AWS Web Application Firewall protects web applications from common web exploits like SQL injection, cross-site scripting (XSS), and Layer 7 DDoS. You create web ACLs with rules (allow/block/count) based on IP addresses, geographic location, request size, and string/regex patterns. WAF integrates with ALB, CloudFront, API Gateway, and AppSync.

Q 127**Interview Question**

What is AWS Shield?

Answer:

AWS Shield provides protection against Distributed Denial of Service (DDoS) attacks. Shield Standard is free and automatically protects all AWS customers from most common, frequently occurring network and transport layer attacks. Shield Advanced (paid) provides enhanced protection against larger attacks, 24/7 access to the DDoS Response Team, and cost protection.

Q 128**Interview Question**

What is AWS GuardDuty?

Answer:

Amazon GuardDuty is a threat detection service that continuously monitors your AWS accounts for malicious activity and unauthorized behavior. It analyzes CloudTrail, VPC Flow Logs, DNS logs, and S3 data events using machine learning and threat intelligence feeds. GuardDuty generates findings categorized by severity — requiring no agents or additional infrastructure.

Q 129**Interview Question**

What is AWS Macie?

Answer:

Amazon Macie is a fully managed data security service that uses ML to discover and protect sensitive data in S3. It automatically identifies PII (Personally Identifiable Information), financial data, health data, and intellectual property. Macie generates alerts when sensitive data is found in unexpected locations or when bucket configurations pose security risks.

Q 130**Interview Question**

What is AWS Security Hub?

Answer:

AWS Security Hub provides a central view of your security posture across AWS accounts. It aggregates, organizes, and prioritizes security findings from GuardDuty, Inspector, Macie, Firewall Manager, IAM Access Analyzer, and third-party tools. Security Hub checks your environment against security standards like CIS AWS Foundations Benchmark and AWS Foundational Security Best Practices.

Q 131**Interview Question**

What is AWS Inspector?

Answer:

Amazon Inspector is an automated vulnerability management service that continuously scans EC2 instances and container images in ECR for software vulnerabilities and unintended network exposure. It uses the Common Vulnerabilities and Exposures (CVE) database and integrates with Systems Manager to assess patching status — providing risk scores for findings.

Q 132**Interview Question****What is VPC Flow Logs?****Answer:**

VPC Flow Logs capture information about IP traffic going to and from network interfaces in your VPC. Log records include source/destination IPs, ports, protocol, bytes transferred, and accept/reject status. Flow logs can be published to CloudWatch Logs or S3. They are essential for security analysis, troubleshooting connectivity issues, and network monitoring.

Q 133**Interview Question****What is AWS Secrets Manager?****Answer:**

AWS Secrets Manager helps you protect access to applications, services, and IT resources by storing and automatically rotating secrets like database credentials, API keys, and OAuth tokens. Unlike Parameter Store SecureString, Secrets Manager has built-in rotation support — it automatically rotates RDS credentials on a schedule you define, reducing the risk of credential exposure.

Q 134**Interview Question****What is IAM Access Analyzer?****Answer:**

IAM Access Analyzer helps you identify resources that are shared with external entities — like S3 buckets or KMS keys shared with accounts outside your organization. It uses automated reasoning (provable security) to identify these access paths. It also validates IAM policies against best practices and checks whether policies grant intended permissions.

Q 135**Interview Question****What is AWS Certificate Manager (ACM)?****Answer:**

ACM lets you provision, manage, and deploy TLS/SSL certificates for use with AWS services like CloudFront, ALB, and API Gateway. ACM certificates are free for public certificates and auto-renew before expiration. Private certificates for internal services use ACM Private CA. Using ACM eliminates the manual certificate procurement and renewal process.

Q 136**Interview Question**

What is a Security Group?

Answer:

A Security Group acts as a virtual firewall for EC2 instances at the instance network interface level. Rules control inbound and outbound traffic based on protocol, port, and source/destination (IP ranges or other security groups). Security Groups are stateful — if inbound traffic is allowed, the response is automatically allowed. You can only add allow rules, not deny.

Q 137**Interview Question**

What is AWS Organizations?

Answer:

AWS Organizations helps you centrally manage and govern multiple AWS accounts. You can group accounts into Organizational Units (OUs), apply Service Control Policies (SCPs) to set permission guardrails, use consolidated billing, and enable AWS services across all accounts simultaneously. It is the foundation for multi-account AWS environments following the Landing Zone pattern.

Q 138**Interview Question**

What is a Service Control Policy (SCP)?

Answer:

SCPs are a type of organizational policy in AWS Organizations that sets the maximum permissions for accounts within an OU or the entire organization. SCPs do not grant permissions themselves — they restrict what IAM policies in member accounts can allow. For example, an SCP can prevent any account in the Org from deleting CloudTrail or leaving the organization.

Q 139**Interview Question**

What is AWS Firewall Manager?

Answer:

AWS Firewall Manager is a security management service to centrally configure and manage firewall rules across all accounts in your AWS Organization. You can deploy WAF rules, Shield Advanced protections, Security Groups, and Network Firewall policies from a single place — ensuring consistent security posture without individually managing each account.

Q 140**Interview Question**

What is Amazon Cognito?

Answer:

Amazon Cognito provides authentication, authorization, and user management for web and mobile applications. User Pools handle user registration, sign-in, and token vending. Identity Pools provide temporary AWS credentials to users (including those authenticated via third-party IdPs) to access AWS services directly. It supports OAuth, SAML, and OpenID Connect.

Q 141**Interview Question**

What is AWS RAM (Resource Access Manager)?

Answer:

AWS RAM lets you share AWS resources with other accounts in your organization or with specific AWS accounts — without creating duplicate resources. Shareable resources include VPC subnets, Route 53 Resolver rules, Transit Gateway, License Manager configurations, and more. Sharing subnets is a common pattern for centralizing network management in multi-account environments.

Q 142**Interview Question**

What is an EC2 Instance Profile?

Answer:

An Instance Profile is a container for an IAM role that you can attach to an EC2 instance. The instance assumes this role and applications running on the instance can call AWS APIs using the role's permissions, with temporary credentials automatically provided through the Instance Metadata Service (IMDS). This eliminates the need for long-term access keys on EC2.

Q 143**Interview Question****What is IMDSv2?****Answer:**

IMDSv2 (Instance Metadata Service v2) is a session-oriented method for accessing EC2 instance metadata that requires a session token. Unlike IMDSv1 (simple GET requests), IMDSv2 requires a PUT request to get a token first, protecting against certain SSRF (Server-Side Request Forgery) attacks. AWS recommends requiring IMDSv2 on all instances via instance metadata options.

Q 144**Interview Question****What is AWS Network Firewall?****Answer:**

AWS Network Firewall is a managed, stateful network firewall and intrusion detection/prevention service for VPCs. It provides fine-grained control over network traffic using stateful and stateless rules — including domain name filtering, protocol detection, and intrusion prevention rules (Suricata-compatible). It integrates with Firewall Manager for centralized policy management.

Q 145**Interview Question****What is S3 Server-Side Encryption?****Answer:**

S3 SSE encrypts objects at rest. Options include: SSE-S3 (AWS manages keys, AES-256), SSE-KMS (uses KMS keys, audit trail, key rotation, additional control), and SSE-C (customer provides encryption keys, AWS does the encryption/decryption but never stores the key). Client-side encryption is also possible where you encrypt before uploading to S3.

Q 146**Interview Question****What is AWS CloudHSM?****Answer:**

AWS CloudHSM provides dedicated Hardware Security Modules in the AWS cloud for generating and storing cryptographic keys. Unlike KMS (multi-tenant, AWS-managed keys), CloudHSM is single-tenant and you have full control of the HSM. It is used for regulatory requirements mandating dedicated HSM hardware, or for applications needing custom cryptographic processing.

Q 147**Interview Question**

What is AWS Config Rules?

Answer:

Config Rules evaluate whether your AWS resources comply with your organization's policies. AWS provides managed rules (like requiring MFA on root, checking S3 bucket versioning, ensuring EC2 instances use approved AMIs). You can also create custom rules using Lambda. Non-compliant resources are flagged, and you can set up auto-remediation using Systems Manager Automation.

Q 148**Interview Question**

What is Amazon Detective?

Answer:

Amazon Detective helps you analyze, investigate, and quickly identify the root cause of potential security issues or suspicious activities. It automatically collects log data from CloudTrail, VPC Flow Logs, and GuardDuty findings and uses ML to build an interactive behavior graph. It is the investigation tool that complements GuardDuty's detection capabilities.

Q 149**Interview Question**

What is IAM Permission Boundary?

Answer:

A Permissions Boundary is an advanced IAM feature that sets the maximum permissions an IAM entity (user or role) can have, regardless of what identity-based policies grant. It is used to delegate IAM permissions to developers safely — a developer can create roles but only up to the permissions defined in the boundary — preventing privilege escalation.

Q 150**Interview Question**

What is AWS Audit Manager?**Answer:**

AWS Audit Manager helps you continuously audit your AWS usage to simplify risk assessment and compliance with regulations and industry standards like HIPAA, GDPR, PCI DSS, and SOC 2. It automatically collects evidence from AWS services, maps it to control requirements, and generates audit-ready assessment reports — reducing the manual effort of compliance audits.

Section 6: Networking (Q151-Q180)

Q 151**Interview Question**

What is a VPC?

Answer:

A Virtual Private Cloud is a logically isolated section of the AWS cloud where you launch AWS resources in a virtual network you define. You control the IP address range (CIDR block), subnets, route tables, internet gateways, and network access control lists. Each AWS account has a default VPC in each region, but production workloads should use custom VPCs.

Q 152**Interview Question**

What are Public and Private Subnets?

Answer:

A public subnet has a route to an Internet Gateway, allowing instances with public IPs to communicate with the internet. A private subnet does not have a direct route to the internet. Instances in private subnets access the internet through a NAT Gateway or NAT Instance. Best practice is to place application and database servers in private subnets for security.

Q 153**Interview Question**

What is an Internet Gateway?

Answer:

An Internet Gateway is a horizontally scaled, redundant, highly available VPC component that enables communication between instances in your VPC and the internet. It performs one-to-one NAT for instances with public IPs. One Internet Gateway per VPC — you attach it to the VPC and add a route in the subnet route table pointing 0.0.0.0/0 to the IGW.

Q 154**Interview Question**

What is a NAT Gateway?

Answer:

A NAT (Network Address Translation) Gateway allows instances in private subnets to initiate outbound connections to the internet (for software updates, etc.) while preventing the internet from initiating connections to those instances. It is managed by AWS, highly available within an AZ, and scales up to 45 Gbps. Deploy one NAT Gateway per AZ for high availability.

Q 155**Interview Question**

What is VPC Peering?

Answer:

VPC Peering creates a direct network connection between two VPCs, allowing instances to communicate using private IP addresses as if they were in the same network. Peering is not transitive — if VPC A peers with B and B peers with C, A cannot reach C unless directly peered. Peering works across regions and accounts. CIDR ranges must not overlap.

Q 156**Interview Question**

What is AWS Transit Gateway?

Answer:

AWS Transit Gateway is a network transit hub that connects VPCs and on-premises networks through a central gateway. Instead of creating a mesh of VPC peering connections (which becomes complex at scale), all VPCs connect to the Transit Gateway — simplifying network management. It supports multicast, route tables, and inter-region peering.

Q 157**Interview Question**

What is AWS Direct Connect?

Answer:

AWS Direct Connect is a dedicated private network connection between your on-premises data center and AWS. It bypasses the public internet, providing more consistent network performance, lower latency, and reduced bandwidth costs for high-volume data transfer. Direct Connect links can be 1 Gbps, 10 Gbps, or 100 Gbps. Use it with a VPN for redundancy.

Q 158**Interview Question**

What is a VPN in AWS?

Answer:

AWS Site-to-Site VPN creates an encrypted IPsec tunnel between your on-premises network and your AWS VPC over the public internet. Each VPN connection includes two redundant tunnels for high availability. Client VPN provides OpenVPN-based remote access for individual users. VPN is less expensive than Direct Connect but offers variable internet-based performance.

Q 159**Interview Question****What is Route 53?****Answer:**

Amazon Route 53 is a highly available DNS web service. It supports domain registration, DNS routing (translating domain names to IP addresses), and health checking. Route 53 offers multiple routing policies: Simple, Failover, Weighted, Latency, Geolocation, Geoproximity, and Multivalue Answer — enabling sophisticated global traffic management.

Q 160**Interview Question****What is CloudFront?****Answer:**

Amazon CloudFront is a content delivery network (CDN) that delivers data, videos, applications, and APIs globally with low latency. It caches content at 400+ edge locations worldwide. CloudFront integrates with S3, ALB, EC2, Lambda@Edge, and WAF. It supports HTTPS with custom certificates, origin failover, and field-level encryption.

Q 161**Interview Question****What is VPC Endpoint?****Answer:**

A VPC Endpoint allows you to privately connect your VPC to supported AWS services without requiring internet gateway, NAT device, VPN, or Direct Connect. Traffic between your VPC and the service does not leave the Amazon network. Interface Endpoints use private IPs (powered by PrivateLink). Gateway Endpoints support S3 and DynamoDB and are added to route tables.

Q 162**Interview Question**

What is AWS PrivateLink?

Answer:

AWS PrivateLink enables you to access AWS services and your own or third-party services hosted in other VPCs without exposing traffic to the public internet. You create VPC endpoint services in the provider VPC and interface VPC endpoints in the consumer VPC. Traffic flows privately over the AWS network — useful for SaaS applications needing private connectivity.

Q 163**Interview Question**

What are Elastic Network Interfaces (ENI)?

Answer:

An ENI is a virtual network interface that you can attach to an EC2 instance in a VPC. Each instance gets a primary network interface, and you can add secondary ENIs from different subnets (but same AZ). ENIs have a primary private IP, optional secondary private IPs, an Elastic IP, a MAC address, and security groups — enabling complex multi-subnet network architectures.

Q 164**Interview Question**

What is AWS Network Access Control List (NACL)?

Answer:

NACLs are optional, stateless firewall layers at the subnet level. Rules are evaluated in numerical order (lowest first), and the first matching rule wins. NACLs support both allow and deny rules for inbound and outbound traffic. Because they are stateless, you must explicitly allow return traffic. The default NACL allows all traffic in and out.

Q 165**Interview Question**

What is Route 53 Resolver?

Answer:

Route 53 Resolver is a DNS resolver for VPCs. It resolves DNS queries for resources within a VPC (using Route 53 private hosted zones) and forwards queries for on-premises domains to your DNS servers. Resolver endpoints (inbound and outbound) integrate with your on-premises DNS infrastructure for hybrid cloud environments using Direct Connect or VPN.

Q 166**Interview Question****What is AWS Global Accelerator vs CloudFront?****Answer:**

CloudFront is a CDN that caches content at edge locations — ideal for static or cacheable content. Global Accelerator routes TCP and UDP traffic over the AWS global network to the nearest AWS endpoint without caching. CloudFront works at Layer 7 (HTTP/HTTPS); Global Accelerator works at Layers 4/7. Use GA for dynamic, non-cacheable workloads requiring fixed IPs and TCP/UDP.

Q 167**Interview Question****What is Egress-Only Internet Gateway?****Answer:**

An Egress-Only Internet Gateway allows outbound IPv6 communication from instances in a VPC to the internet, while preventing the internet from initiating connections with IPv6 instances. It is the IPv6 equivalent of a NAT Gateway. Unlike NAT Gateway (which is for IPv4 private addresses), this is used specifically when your VPC uses IPv6 addressing.

Q 168**Interview Question****What is VPC Flow Logs retention?****Answer:**

VPC Flow Logs can be delivered to CloudWatch Logs or Amazon S3. For CloudWatch Logs, retention is controlled by the log group's retention policy (1 day to 10 years). For S3, data persists according to S3 lifecycle policies. For long-term storage and cost optimization, S3 is preferred over CloudWatch Logs for flow log archives.

Q 169**Interview Question****What is Reachability Analyzer?****Answer:**

AWS Reachability Analyzer is a network diagnostic tool that lets you verify network connectivity between two resources in your VPC without sending actual traffic. You specify a source and destination (EC2 instances, ENIs, IGWs, etc.), and the tool analyzes your network configuration and identifies blockers like security group rules, NACLs, or route table misconfigurations.

Q 170**Interview Question**

What is Network Manager in AWS?

Answer:

AWS Network Manager provides a central dashboard to manage, monitor, and troubleshoot your global network across AWS and on-premises locations. It visualizes your network topology including Transit Gateways, VPCs, VPN connections, Direct Connect gateways, and SD-WAN devices — giving you a single pane of glass for complex hybrid and multi-region networks.

Q 171**Interview Question**

What is IPv6 in AWS VPC?

Answer:

AWS VPCs support dual-stack (IPv4 and IPv6) configurations. IPv6 CIDR blocks are assigned from Amazon's pool (/56 per VPC, /64 per subnet). All IPv6 addresses in AWS are public and globally unique — there is no private IPv6 (use Egress-Only IGW to prevent inbound). IPv6 enables direct internet connectivity and eliminates NAT complexity for compatible applications.

Q 172**Interview Question**

What is a Bastion Host?

Answer:

A Bastion Host (also called a jump box) is an EC2 instance in a public subnet that you use as a secure gateway to access instances in private subnets via SSH or RDP. Traffic flows: your computer → Bastion (public IP) → private instance (private IP). Modern best practice is to use AWS Systems Manager Session Manager instead, which eliminates the need for bastion hosts entirely.

Q 173**Interview Question**

What is Route 53 Weighted Routing?

Answer:

Weighted Routing lets you route traffic to multiple resources in proportions you specify. For example, you might send 80% of traffic to your current version and 20% to a new version for A/B testing or gradual blue/green migration. Weights range from 0 to 255. Setting a weight of 0 for a record stops traffic to it while keeping the record in place.

Q 174**Interview Question**

What is VPC CIDR?

Answer:

A VPC CIDR (Classless Inter-Domain Routing) block is the IP address range for your VPC (e.g., 10.0.0.0/16). It must be between /16 and /28. You can add secondary CIDR blocks after creation but cannot remove or modify the primary block. Choose a CIDR that does not overlap with your on-premises network or other VPCs you plan to peer with.

Q 175**Interview Question**

What is AWS Marketplace?

Answer:

AWS Marketplace is a digital catalog of thousands of software listings from independent software vendors (ISVs) that run on AWS. Products include AMIs, CloudFormation templates, SaaS products, and containers. For SysOps, it is useful for quickly deploying pre-configured network appliances, security tools, monitoring software, and databases without manual setup.

Q 176**Interview Question**

What is an Elastic IP vs Public IP?

Answer:

A regular public IP address is dynamically assigned when an EC2 instance starts and released when it stops — it changes on every restart. An Elastic IP is a static public IPv4 address you allocate to your account. It remains associated with your account until you release it, allowing you to remap it to different instances for quick failover without DNS propagation delays.

Q 177

Interview Question

What is AWS CloudFront Origin Shield?**Answer:**

Origin Shield is an additional caching layer in CloudFront that sits between edge locations and your origin. It reduces the load on your origin by consolidating requests — all cache misses from edge locations go to Origin Shield first before hitting your origin. This significantly reduces origin requests, improving cache hit ratios and reducing origin infrastructure costs.

Q 178

Interview Question

What are VPC Reserved IPs?**Answer:**

AWS reserves 5 IP addresses in each subnet CIDR block — the first four and the last: .0 (network address), .1 (default gateway/VPC router), .2 (AWS DNS), .3 (reserved for future use), and .255 (broadcast, not supported). For a /24 subnet with 256 addresses, you get 251 usable IPs. Plan your CIDR sizes accounting for these reserved addresses.

Q 179

Interview Question

What is Route 53 Failover Routing?**Answer:**

Failover Routing configures an active-passive setup where Route 53 routes traffic to a primary resource when it is healthy and automatically fails over to a secondary resource when the primary fails health checks. It is typically used with a primary EC2 in us-east-1 and a standby S3 static website or instance in another region as the failover target.

Q 180

Interview Question

What is AWS Wavelength?**Answer:**

AWS Wavelength embeds AWS compute and storage services at the edge of 5G networks, inside telecommunications carriers' data centers. Applications hosted in Wavelength Zones can serve mobile users with single-digit millisecond latency over the 5G network. It is designed for use cases like live video streaming, AR/VR, connected vehicles, and real-time gaming.

Section 7: Cost Optimization (Q181-Q210)

Q 181**Interview Question**

What is AWS Cost Explorer?

Answer:

AWS Cost Explorer is a tool that visualizes, understands, and manages your AWS costs and usage over time. It provides preconfigured views for monthly costs by service, costs by account, and RI coverage. You can create custom charts, filter by service/tag/account, and get cost forecasts. It also offers Rightsizing Recommendations to identify over-provisioned EC2 instances.

Q 182**Interview Question**

What are Reserved Instances (RI)?

Answer:

Reserved Instances provide a significant discount (up to 72%) compared to On-Demand pricing in exchange for a 1-year or 3-year commitment. Standard RIs offer the highest discount but least flexibility. Convertible RIs allow you to change instance families, OS, and tenancy. You pay upfront (all, partial, or no upfront). RIs apply automatically to matching On-Demand usage.

Q 183**Interview Question**

What are Savings Plans?

Answer:

Savings Plans offer up to 66% savings over On-Demand in exchange for a commitment to a consistent amount of compute usage (\$/hour) for 1 or 3 years. Compute Savings Plans apply automatically across EC2, Fargate, and Lambda regardless of instance family, region, OS, or tenancy — more flexible than RIs. EC2 Instance Savings Plans apply to specific instance families in a region.

Q 184**Interview Question**

What is AWS Trusted Advisor?

Answer:

AWS Trusted Advisor is an online tool that provides real-time guidance to help you provision resources following AWS best practices. It checks across five categories: Cost Optimization (idle resources, underutilized instances), Performance, Security (open security groups, MFA), Fault Tolerance (backups, multi-AZ), and Service Limits. Full checks require Business or Enterprise Support.

Q 185**Interview Question**

What is AWS Cost and Usage Report (CUR)?

Answer:

The Cost and Usage Report is the most comprehensive billing dataset available in AWS. It provides hourly or daily line items for every AWS service you use, including resource IDs, usage amounts, rates, tags, and Reserved Instance/Savings Plans amortization. CUR is published to S3 and can be analyzed with Athena, QuickSight, or third-party tools.

Q 186**Interview Question**

What is an EC2 Spot Fleet?

Answer:

A Spot Fleet is a collection of Spot Instances and optionally On-Demand Instances that attempts to meet your target capacity within the price constraints you set. You define multiple launch pools (instance type + AZ + OS combinations), and the fleet automatically diversifies across them to maintain capacity even when Spot prices rise in one pool — maximizing availability at low cost.

Q 187**Interview Question**

What is AWS Compute Optimizer?

Answer:

AWS Compute Optimizer uses ML to analyze CloudWatch utilization metrics from your EC2 instances, Auto Scaling groups, Lambda functions, and EBS volumes. It recommends optimal AWS resources to reduce costs and improve performance — for example, suggesting a smaller instance type if CPU utilization consistently remains below 10%.

Q 188**Interview Question**

What is AWS Budgets?

Answer:

AWS Budgets lets you set custom budgets for costs, usage, Savings Plans utilization, and RI coverage. When actual or forecasted values exceed your threshold, Budgets sends alerts via email or SNS. You can also set up Budget Actions that automatically apply IAM policies, SCP restrictions, or target EC2/RDS instances when a budget threshold is breached.

Q 189**Interview Question****What are EC2 On-Demand Capacity Reservations?****Answer:**

On-Demand Capacity Reservations reserve EC2 capacity for a specific instance type, AZ, and platform without a long-term commitment. Unlike Reserved Instances, they do not provide a discount — you pay On-Demand rates. But they guarantee capacity is available when needed, which is critical for DR scenarios where you must launch instances immediately.

Q 190**Interview Question****What is S3 Storage Class Analysis?****Answer:**

S3 Storage Class Analysis observes data access patterns over time (typically 30+ days) and provides recommendations on which objects could be transitioned to S3 Standard-IA for cost savings. It generates reports you can view in the S3 console or export to CSV. This data helps you make informed decisions when configuring lifecycle policies.

Q 191**Interview Question****What is AWS Cost Allocation Tags?****Answer:**

Cost Allocation Tags let you label AWS resources with key-value pairs (e.g., Environment:Production, Team:DataEngineering) to track costs by project, team, or cost center. You must activate tags as cost allocation tags in the Billing console. They then appear as columns in Cost Explorer and CUR, enabling granular cost attribution and chargeback.

Q 192**Interview Question**

What is Amazon EC2 rightsizing?

Answer:

Rightsizing is the process of matching your EC2 instance types and sizes to workload requirements to reduce costs. Tools like Cost Explorer Rightsizing Recommendations and Compute Optimizer analyze CloudWatch metrics and identify idle or over-provisioned instances. You should analyze at least 2 weeks of data and test in non-production before resizing production workloads.

Q 193**Interview Question**

What is AWS Data Transfer costs?

Answer:

Data transfer pricing: Inbound to AWS is free. Outbound to the internet is charged (varies by region, tiered pricing). Transfer between AWS services in the same region is often free or minimal. Cross-region transfer between AWS services is charged. Use VPC endpoints to avoid data transfer charges for S3 and DynamoDB. CloudFront reduces origin data transfer costs.

Q 194**Interview Question**

What is Spot Instance interruption?

Answer:

AWS can interrupt (terminate, stop, or hibernate) Spot Instances with a 2-minute warning notification via EC2 instance metadata and EventBridge. To build resilient Spot-based architectures: save work frequently, use checkpointing, design applications to be stateless, use mixed fleet strategies with On-Demand fallback, and use Spot-friendly services like AWS Batch or ECS with Fargate Spot.

Q 195**Interview Question**

What is AWS Free Tier?

Answer:

AWS Free Tier provides limited free usage of AWS services for 12 months after account creation (e.g., 750 hours/month of t2.micro EC2, 5 GB S3 storage, 25 GB DynamoDB) plus always-free offers (Lambda: 1M requests/month, 400,000 GB-seconds compute time). Staying within free tier limits requires monitoring with AWS Budgets and Cost Explorer.

Q 196**Interview Question****What is S3 Requester Pays and its cost impact?****Answer:**

With Requester Pays enabled, data transfer and request costs are charged to the requester rather than the bucket owner. This is useful for distributing large public datasets where you want to share data without absorbing bandwidth costs. The bucket owner still pays for storage. Requesters must provide authentication credentials — anonymous access is not allowed.

Q 197**Interview Question****What is EC2 Instance Savings Plan vs Compute Savings Plan?****Answer:**

EC2 Instance Savings Plans commit to a specific instance family in a region (e.g., m5 in us-east-1) and provide up to 72% savings — similar to Standard RIs but more flexible (any size/OS in the family). Compute Savings Plans apply to any instance family, region, OS, and size across EC2, Fargate, and Lambda — offering up to 66% savings with maximum flexibility.

Q 198**Interview Question****What is AWS Cost Anomaly Detection?****Answer:**

AWS Cost Anomaly Detection uses ML to identify unusual spending patterns in your AWS account and send alerts. You create monitors (for individual services, member accounts, cost categories, or specific tags) and set thresholds for alert notifications. It can detect both sudden spikes and gradual cost increases, helping you catch billing issues early.

Q 199**Interview Question****What is Consolidated Billing?****Answer:**

Consolidated Billing in AWS Organizations combines the usage across all accounts to potentially lower the overall cost through volume pricing discounts (more usage = lower per-unit rates for services like S3 and data transfer). Reserved Instance and Savings Plans discounts are shared across accounts automatically. One bill for all accounts simplifies financial management.

Q 200**Interview Question**

What is EBS GP2 vs GP3 cost difference?

Answer:

GP3 is 20% cheaper than GP2 per GB-month (\$0.08 vs \$0.10 per GB-month in us-east-1). Additionally, GP3 provides 3,000 IOPS and 125 MB/s throughput free. GP2 IOPS scales with size (3 IOPS/GB). For workloads needing more than 3,000 IOPS on GP2, GP3 allows increasing IOPS independently (up to 16,000) at lower cost. Always prefer GP3 for new volumes.

Q 201**Interview Question**

What is AWS Pricing Calculator?

Answer:

AWS Pricing Calculator is a free web-based planning tool to estimate the cost of your AWS use cases before deploying. You add services, configure their settings (instance type, hours, data transfer, etc.), and the calculator shows the monthly estimate. You can create multiple scenarios, save them, and share them — useful for project proposals and budgeting.

Q 202**Interview Question**

What is EC2 Scheduled Reserved Instances?

Answer:

Scheduled Reserved Instances allow you to purchase capacity reservations that recur on a schedule (daily, weekly, or monthly) for a 1-year term. You specify the time window, instance type, and AZ. This is ideal for predictable, time-boxed workloads like daily batch jobs or weekly report generation that run for a few hours at a time.

Q 203**Interview Question**

What is Amazon S3 cost optimization strategies?

Answer:

Key strategies: Use lifecycle policies to transition to cheaper storage classes (IA, Glacier). Enable Intelligent-Tiering for unpredictable access patterns. Review and clean up incomplete multipart uploads. Use S3 Storage Lens to identify buckets with low activity. Enable Requester Pays for shared datasets. Delete non-current versions if versioning is not needed for compliance.

Q 204**Interview Question****What is AWS Cost Categories?****Answer:**

AWS Cost Categories let you map your AWS costs into meaningful business categories using custom rules. Rules can match by accounts, tags, services, or charge types. For example, create a Development category that aggregates costs from all accounts tagged Environment:Dev. Cost Categories appear in Cost Explorer, Budgets, and CUR for better cost visibility and reporting.

Q 205**Interview Question****What is RDS Reserved Instances?****Answer:**

RDS Reserved Instances provide up to 69% discount over On-Demand RDS pricing in exchange for a 1-year or 3-year commitment. You specify the database engine, instance class, Multi-AZ, and region. Unlike EC2 RIs, RDS RIs are not convertible — you must know your database requirements in advance. They are applied automatically to matching running RDS instances.

Q 206**Interview Question****What is Lambda pricing?****Answer:**

Lambda charges based on two dimensions: number of requests (\$0.20 per 1M requests after the free tier) and duration (rounded to the nearest 1ms, priced per GB-second of memory configured × duration). The free tier includes 1M requests and 400,000 GB-seconds per month permanently. Using the right memory setting and optimizing function duration directly impacts cost.

Q 207**Interview Question****What is AWS Cost Explorer Rightsizing Recommendations?****Answer:**

This feature within Cost Explorer analyzes CloudWatch CPU utilization metrics from EC2 instances and recommends rightsizing actions — either terminating idle instances or downsizing over-provisioned ones. It shows estimated monthly savings for each recommendation. You can filter by tag, account, or region and export recommendations for review.

Q 208**Interview Question****What is Spot Instance pricing?****Answer:**

Spot Instance prices fluctuate based on supply and demand but are typically 70-90% cheaper than On-Demand prices. Prices vary by instance type, AZ, and region. AWS no longer uses bidding — you specify your maximum price (default is On-Demand price), and you pay the current Spot price when running. Spot Instances are best for flexible, fault-tolerant batch workloads.

Q 209**Interview Question****What is AWS Purchase Order management?****Answer:**

AWS allows enterprise customers to associate purchase orders with their billing account. You create POs with invoice and line item configurations so that invoices reflect your PO numbers — essential for enterprise procurement workflows. PO amounts can be set to trigger notifications or alerts when thresholds are reached, aiding financial management.

Q 210**Interview Question****What is EBS volume cost optimization?****Answer:**

Identify and delete unattached EBS volumes (a common source of waste). Use Snapshots Lifecycle Manager to automate and clean up old snapshots. Migrate from GP2 to GP3 for cost savings. Right-size volumes (reduce size if disk utilization is consistently low). Archive rarely accessed snapshots using EBS Snapshot Archive. Use CloudWatch disk utilization metrics from the CW agent to inform decisions.

Section 8: Automation & Configuration Management (Q211-Q240)

Q 211**Interview Question**

What is AWS Systems Manager Automation?

Answer:

SSM Automation simplifies common maintenance and deployment tasks through Automation runbooks (SSM Documents of type Automation). You can automate tasks like creating AMIs, patching instances, starting/stopping EC2, or managing RDS snapshots. Runbooks can be triggered manually, on a schedule via EventBridge, or as a remediation action in Config Rules — enabling operational workflows without manual intervention.

Q 212**Interview Question**

What is AWS Systems Manager Run Command?

Answer:

Run Command lets you remotely and securely run scripts on EC2 instances and on-premises servers without SSH or RDP access. You select target instances by instance IDs or resource tags, choose a document (like AWS-RunShellScript), provide the command, and Run Command executes it on all targets in parallel. Results and status are available in the console or S3.

Q 213**Interview Question**

What is AWS Systems Manager Patch Manager?

Answer:

Patch Manager automates the process of patching managed nodes with security-related and other updates. You define patch baselines (which patches are approved), create maintenance windows (scheduled patching time), and configure patch groups to apply specific baselines to specific instances. It works on both EC2 and on-premises servers with the SSM agent installed.

Q 214**Interview Question**

What is AWS Systems Manager Session Manager?

Answer:

Session Manager provides secure, browser-based interactive shell access to EC2 instances and on-premises servers without needing SSH keys, open inbound ports, or bastion hosts. Sessions are logged and audited in CloudTrail and S3. Access is controlled via IAM policies. It works through the SSM agent and eliminates the security overhead of managing SSH keys.

Q 215**Interview Question**

What is an SSM Document?

Answer:

An SSM Document (also called a runbook or playbook) is a JSON or YAML document that defines actions to be performed on managed instances. Types include Command documents (used by Run Command and State Manager), Automation documents (for multi-step workflows), Session documents (Session Manager configurations), and Policy documents (for Inventory). AWS provides hundreds of pre-built documents.

Q 216**Interview Question**

What is AWS Lambda?

Answer:

AWS Lambda is a serverless compute service that runs your code in response to events without provisioning or managing servers. You upload code, configure triggers (API Gateway, S3, DynamoDB, SNS, EventBridge, etc.), and Lambda automatically scales. You pay only for the compute time consumed (per 1ms increments). Supported runtimes include Python, Node.js, Java, Go, .NET, and Ruby.

Q 217**Interview Question**

What is AWS Step Functions?

Answer:

AWS Step Functions is a serverless workflow service that coordinates distributed application components using visual workflows called state machines. You define states (tasks, choices, parallel, wait, success, fail) in Amazon States Language. Step Functions manages state, error handling, and retries — ideal for multi-step workflows like order processing, ETL pipelines, and approval flows.

Q 218**Interview Question**

What is AWS CloudFormation Helper Scripts?

Answer:

CloudFormation helper scripts (cfn-init, cfn-signal, cfn-hup, cfn-get-metadata) are Python scripts that you install on EC2 instances to communicate with CloudFormation. cfn-init reads metadata from the template and applies configuration (packages, files, services). cfn-signal sends success/failure signals to WaitConditions. cfn-hup detects and applies metadata changes without stack updates.

Q 219**Interview Question**

What is EventBridge (CloudWatch Events)?

Answer:

Amazon EventBridge is a serverless event bus that routes events from AWS services, custom applications, and SaaS partners to targets like Lambda, SNS, SQS, and Step Functions. Rules match event patterns or run on a schedule (cron/rate). It enables event-driven architectures and automated operational responses — like triggering a Lambda to clean up resources when a CloudFormation stack fails.

Q 220**Interview Question**

What is AWS Config Remediation?

Answer:

AWS Config Remediation automatically applies corrective actions when a Config Rule finds a non-compliant resource. You configure a remediation action (linked to an SSM Automation document) that runs when a rule evaluates a resource as non-compliant. You can choose manual remediation (triggered by an operator) or automatic remediation (runs immediately on detection).

Q 221**Interview Question**

What is AWS Systems Manager State Manager?

Answer:

State Manager is a secure and scalable configuration management service that automates the process of keeping your EC2 instances and hybrid infrastructure in a defined, consistent state. You create associations that link a document to a target and define a schedule. For example, run a script every hour to ensure CloudWatch agent is running on all instances.

Q 222**Interview Question**

What is AWS Systems Manager Inventory?

Answer:

SSM Inventory collects metadata from your managed instances including installed applications, network configuration, Windows Update history, running services, and more. You can view this data in Systems Manager or sync it to S3 and query it with Athena. Inventory helps you understand what software and configurations exist across your entire fleet.

Q 223**Interview Question**

What is AWS CodeBuild?

Answer:

AWS CodeBuild is a fully managed continuous integration service that compiles source code, runs tests, and produces software packages. You define a build specification (buildspec.yml) with phases (install, pre-build, build, post-build) and commands. CodeBuild scales automatically and eliminates the need to manage build servers — integrating with CodePipeline for CI/CD.

Q 224**Interview Question**

What is the AWS CDK?

Answer:

The AWS Cloud Development Kit lets you define cloud infrastructure using familiar programming languages like TypeScript, Python, Java, and C#. CDK synthesizes your code into CloudFormation templates. It provides high-level constructs (L2 and L3) that abstract common patterns — reducing boilerplate and enabling developers to use IDE features like autocomplete and type checking.

Q 225**Interview Question**

What is AWS SAM (Serverless Application Model)?**Answer:**

AWS SAM is an open-source framework for building serverless applications. It extends CloudFormation with simplified syntax for defining Lambda functions, API Gateways, DynamoDB tables, and event source mappings. SAM CLI provides local testing and debugging capabilities. SAM templates are transformed into CloudFormation templates for deployment — simplifying serverless IaC.

Q 226**Interview Question****What is AWS Elastic Container Service (ECS)?****Answer:**

Amazon ECS is a fully managed container orchestration service for running Docker containers on AWS. You define tasks (container definitions) and services (maintain desired count, integrate with load balancers). ECS runs containers on EC2 instances (EC2 launch type) or serverlessly using AWS Fargate. It integrates with ECR, IAM, CloudWatch, and App Mesh.

Q 227**Interview Question****What is AWS Fargate?****Answer:**

AWS Fargate is a serverless compute engine for containers. When using Fargate with ECS or EKS, you do not manage the underlying EC2 instances — you specify CPU and memory requirements, and AWS handles everything else. Fargate eliminates the need to provision and manage servers, letting you focus on application design rather than infrastructure management.

Q 228**Interview Question****What is AWS OpsWorks vs Systems Manager?****Answer:**

OpsWorks (using Chef/Puppet) is for teams with existing Chef/Puppet expertise who need complex configuration management workflows. Systems Manager is AWS-native and does not require Chef/Puppet knowledge — it covers patching, configuration, run command, and inventory through a unified AWS interface. For new AWS-centric workloads, Systems Manager is the preferred choice.

Q 229**Interview Question**

What is AWS Elastic Kubernetes Service (EKS)?

Answer:

Amazon EKS is a managed Kubernetes service that makes it easy to run Kubernetes on AWS without needing to install, operate, or maintain your own Kubernetes control plane. AWS manages the control plane (API servers, etcd). You manage worker nodes (EC2 or Fargate). EKS integrates with IAM, VPC networking, ALB, CloudWatch, and ECR.

Q 230**Interview Question**

What is Terraform vs CloudFormation?

Answer:

Both are IaC tools. CloudFormation is AWS-native, free, tightly integrated with all AWS services, and uses JSON/YAML. Terraform is multi-cloud (works with AWS, Azure, GCP), uses HCL, has a large community, and offers a state file for tracking infrastructure. Organizations with multi-cloud strategies prefer Terraform; AWS-only organizations often prefer CloudFormation for deep service integration.

Q 231**Interview Question**

What is AWS Service Quotas?

Answer:

AWS Service Quotas (formerly Limits) is a centralized service to view and manage your quotas for AWS services. Each service has default quotas (like the maximum number of EC2 instances per region). You can request quota increases directly from the Service Quotas console. CloudWatch can monitor quota usage, and you can set alarms before reaching limits to avoid service disruptions.

Q 232**Interview Question**

What is Amazon Managed Workflows for Apache Airflow (MWAA)?**Answer:**

Amazon MWAA is a managed service for Apache Airflow — an open-source tool for orchestrating complex data workflows as directed acyclic graphs (DAGs). AWS handles provisioning, scaling, availability, and maintenance of the Airflow environment. It integrates with S3 (for DAG storage), IAM, CloudWatch, and VPC — ideal for data engineering and ETL workflow automation.

Q 233**Interview Question****What is AWS CloudFormation WaitCondition?****Answer:**

WaitConditions pause a CloudFormation stack creation/update until a signal is received. The signal comes from an EC2 instance (via cfn-signal) after completing initialization steps like software installation. If the signal is not received within the timeout, the stack fails and rolls back. WaitConditions ensure CloudFormation waits for instances to be fully ready before proceeding.

Q 234**Interview Question****What is AWS AppRunner?****Answer:**

AWS App Runner is a fully managed service that makes it easy to deploy containerized web applications and APIs from a container registry or a source code repository. App Runner automatically builds, deploys, scales, and manages the infrastructure — requiring no container orchestration knowledge. It is the simplest way to deploy containerized apps on AWS.

Q 235**Interview Question****What is AWS Batch?****Answer:**

AWS Batch is a fully managed service for running batch computing workloads at any scale. You define job queues, compute environments (EC2 or Fargate, supporting Spot Instances), and job definitions. Batch automatically provisions the right amount and type of compute resources. It is ideal for genomics, financial modeling, simulation, and data transformation pipelines.

Q 236**Interview Question**

What is AWS Glue?

Answer:

AWS Glue is a serverless data integration service for discovering, preparing, and combining data for analytics, ML, and application development. Glue crawlers catalog data sources, the Glue Data Catalog stores metadata, and Glue ETL jobs transform data using auto-generated or custom PySpark/Python scripts. It integrates with S3, RDS, Redshift, and Athena.

Q 237**Interview Question**

What is Amazon EventBridge Pipes?

Answer:

EventBridge Pipes creates point-to-point integrations between event producers and consumers with filtering, enrichment, and transformation. A Pipe connects a source (Kinesis, SQS, DynamoDB stream) to a target (Lambda, Step Functions, API destination) with optional filtering and enrichment middleware — reducing integration code significantly.

Q 238**Interview Question**

What is AWS CloudFormation nested stacks?

Answer:

Nested Stacks allow you to reuse common CloudFormation template patterns by embedding one stack (child) inside another (parent) using the `AWS::CloudFormation::Stack` resource. The parent passes parameters to the child, and the child outputs values back. This promotes modularity — VPC, security, and compute layers can be separate reusable templates composed by a master template.

Q 239**Interview Question**

What is AWS Lambda Layers?**Answer:**

Lambda Layers are archives containing libraries, custom runtimes, or other function dependencies. Instead of packaging dependencies in every function deployment package, you create a layer, publish it, and reference it from multiple functions. Layers reduce deployment package size, enable sharing of common code across functions, and support updating dependencies independently of function code.

Q 240**Interview Question****What is Infrastructure as Code (IaC) and why is it important for SysOps?****Answer:**

IaC means managing and provisioning infrastructure through code rather than manual processes. For SysOps, IaC provides repeatability (same environment every time), version control (track changes in Git), automated testing, faster disaster recovery (redploy entire stacks in minutes), and documentation (the template is the docs). Tools include CloudFormation, CDK, SAM, and Terraform.

Section 9: Database Services (Q241-Q270)

Q 241**Interview Question**

What is Amazon RDS?

Answer:

Amazon Relational Database Service is a managed database service that makes it easy to set up, operate, and scale relational databases in the cloud. RDS supports MySQL, PostgreSQL, MariaDB, Oracle, SQL Server, and Amazon Aurora. AWS handles backups, software patching, automatic failure detection, and recovery — reducing DBA overhead significantly.

Q 242**Interview Question**

What is Amazon Aurora?

Answer:

Amazon Aurora is an AWS-developed relational database compatible with MySQL and PostgreSQL. It is up to 5x faster than standard MySQL and 3x faster than PostgreSQL. Aurora uses a shared, distributed storage system that automatically scales from 10 GB to 128 TB. It replicates data 6 ways across 3 AZs and has features like Aurora Serverless, Global Database, and Multi-Master.

Q 243**Interview Question**

What is Amazon DynamoDB?

Answer:

DynamoDB is a fully managed, serverless, key-value and document NoSQL database. It delivers single-digit millisecond performance at any scale with automatic multi-region replication (Global Tables), automatic scaling, built-in backup, and encryption at rest. It is ideal for high-traffic web applications, gaming leaderboards, IoT, and any workload requiring consistent low-latency access.

Q 244**Interview Question**

What is Amazon RDS Multi-AZ?

Answer:

RDS Multi-AZ maintains a synchronous standby replica in a different AZ. Writes are committed to the primary and standby before returning success. If the primary fails (hardware, OS, or database failure), RDS automatically fails over to the standby — typically within 1-2 minutes. The DNS endpoint remains the same. Multi-AZ is for high availability, not for read scaling.

Q 245**Interview Question**

What is Amazon RDS Read Replica?

Answer:

Read Replicas provide read scalability by creating asynchronous copies of the primary database. Applications that read heavily can direct read queries to Read Replicas, reducing load on the primary. You can have up to 15 Read Replicas for Aurora and 5 for other RDS engines. Cross-region Read Replicas support geographic distribution and can be promoted to standalone databases.

Q 246**Interview Question**

What is Amazon ElastiCache?

Answer:

ElastiCache is a managed in-memory caching service supporting Redis and Memcached. It is used to reduce database load by caching frequently accessed data, improving application response times. Redis supports persistence, replication, pub/sub, sorted sets, and clustering. Memcached is simpler and multithreaded — suited for simple object caching.

Q 247**Interview Question**

What is Amazon Redshift?

Answer:

Amazon Redshift is a cloud data warehouse service for analyzing large datasets using SQL. It uses columnar storage, massively parallel processing, and result caching to deliver fast query performance. Redshift Spectrum can query data directly in S3 without loading it. Redshift Serverless scales automatically. It integrates with S3, Glue, QuickSight, and SageMaker.

Q 248**Interview Question**

What is DynamoDB TTL (Time To Live)?**Answer:**

DynamoDB TTL automatically deletes items after a specified time at no extra cost. You define a TTL attribute on your table (containing a Unix epoch timestamp). DynamoDB checks TTL every few minutes and deletes expired items in the background. This is useful for session data, event logs, and temporary records that should be cleaned up automatically without incurring write capacity.

Q 249**Interview Question****What is Amazon Aurora Serverless?****Answer:**

Aurora Serverless automatically starts up, shuts down, and scales capacity up or down based on application needs. You specify minimum and maximum Aurora Capacity Units (ACUs). When the database is idle, it scales to zero (in v1) or minimum capacity (in v2), and you pay only for capacity used. It is ideal for infrequent, variable, or unpredictable workloads.

Q 250**Interview Question****What is RDS Automated Backups?****Answer:**

RDS automatically creates daily backups of your DB instance and retains transaction logs, enabling point-in-time recovery (PITR) to any second during your retention window (1-35 days). Backups are stored in S3. The backup window should be set during low-traffic periods. When you delete an RDS instance, automated backups are deleted unless you create a final snapshot.

Q 251**Interview Question****What is Amazon Neptune?****Answer:**

Amazon Neptune is a fully managed graph database service that supports Property Graph (Apache TinkerPop Gremlin) and RDF/SPARQL graph models. It is optimized for storing and querying billions of relationships with millisecond latency. Neptune is ideal for social networking, recommendation engines, fraud detection, knowledge graphs, and network security analysis.

Q 252**Interview Question**

What is DynamoDB Global Tables?

Answer:

DynamoDB Global Tables is a multi-region, multi-active database. A Global Table replicates data automatically across multiple AWS regions, and applications can write to any region's replica. Conflicts are resolved using last-writer-wins. Global Tables provide low-latency access for globally distributed applications and built-in disaster recovery across regions.

Q 253**Interview Question**

What is RDS Parameter Groups?

Answer:

RDS Parameter Groups are containers for engine configuration values (like `max_connections`, `innodb_buffer_pool_size`, `log_bin_trust_function_creators`). You create a custom parameter group, modify parameters, and associate it with your DB instance. Dynamic parameters take effect immediately; static parameters require a DB restart. AWS provides default parameter groups that cannot be modified.

Q 254**Interview Question**

What is Amazon DocumentDB?

Answer:

Amazon DocumentDB is a fully managed JSON document database service compatible with MongoDB. It is designed for workloads that use MongoDB-compatible APIs. DocumentDB decouples compute from storage, stores 6 copies of data across 3 AZs, supports up to 15 low-latency read replicas, and automatically heals storage. It is ideal for content management, catalogs, and user profiles.

Q 255**Interview Question**

What is Amazon Keyspaces (for Apache Cassandra)?

Answer:

Amazon Keyspaces is a managed, serverless, Cassandra-compatible database. It is automatically scalable, highly available (replicates data 3 times across multiple AZs), and enterprise-grade. Existing Cassandra workloads can migrate with minimal application changes using the Cassandra Query Language (CQL). There are no servers to manage or Cassandra software to install.

Q 256**Interview Question**

What is RDS Proxy?

Answer:

Amazon RDS Proxy is a fully managed, highly available database proxy. It pools and shares database connections between applications, reducing the burden of managing thousands of simultaneous connections from serverless applications (Lambda). RDS Proxy improves failover times (by up to 66%) by reusing existing connections and integrates with Secrets Manager for credential management.

Q 257**Interview Question**

What is DynamoDB Streams?

Answer:

DynamoDB Streams captures a time-ordered sequence of item-level modifications in a DynamoDB table. Each stream record captures what changed and how (INSERT, MODIFY, REMOVE) along with before/after images. Streams integrate with Lambda for real-time triggers (e.g., send notification when new order created), and Kinesis Data Streams for analytics pipelines.

Q 258**Interview Question**

What is Amazon MemoryDB for Redis?

Answer:

Amazon MemoryDB for Redis is a Redis-compatible, durable, in-memory database. Unlike ElastiCache (which is a cache that can lose data on failure), MemoryDB stores data durably across multiple AZs with microsecond read and single-digit millisecond write latency. It is designed for applications that need a fast primary database, not just a cache layer.

Q 259

Interview Question

What is RDS Enhanced Monitoring?**Answer:**

Enhanced Monitoring collects metrics from an agent running on the EC2 instance hosting the RDS database. Unlike standard CloudWatch RDS metrics (hypervisor level), Enhanced Monitoring provides OS-level metrics: CPU usage by process, memory, disk I/O, file system usage, and network throughput. It publishes data to CloudWatch Logs at 1-60 second granularity.

Q 260

Interview Question

What is Amazon Timestream?**Answer:**

Amazon Timestream is a fully managed time series database for IoT and operational applications. It stores trillions of events per day and automatically manages data retention with tiered storage (in-memory for recent data, magnetic for historical data). Its built-in time series analytics functions make it ideal for monitoring, IoT sensor data, and application telemetry.

Q 261

Interview Question

What is DynamoDB Accelerator (DAX)?**Answer:**

DAX is a fully managed in-memory caching service specifically for DynamoDB. It delivers up to 10x performance improvement for read-heavy workloads — reducing response times from milliseconds to microseconds. DAX is API-compatible with DynamoDB, requiring minimal application changes. It handles cache invalidation and cluster management automatically.

Q 262

Interview Question

What is RDS Storage Auto Scaling?**Answer:**

RDS Storage Auto Scaling automatically increases storage capacity when free space runs low. You set a Maximum Storage Threshold, and RDS scales storage automatically without downtime when free space falls below 10% and is projected to remain low. This prevents application outages due to running out of database storage. There is no option to scale down storage after scaling up.

Q 263**Interview Question**

What is Amazon QLDB?

Answer:

Amazon QLDB (Quantum Ledger Database) is a fully managed ledger database with a transparent, immutable, and cryptographically verifiable transaction log. It is owned by AWS (centralized, unlike blockchain). QLDB maintains a complete, verifiable history of all application data changes — ideal for systems of record, financial transactions, supply chain, and compliance tracking.

Q 264**Interview Question**

What is the RDS maintenance window?

Answer:

The maintenance window is a weekly time window (e.g., Mon 00:00-01:00 UTC) during which RDS applies OS and database engine patches, including required security patches. If a maintenance is required, RDS applies it during the window even if a deferral is selected. Multi-AZ deployments reduce maintenance impact by patching the standby first, then failing over.

Q 265**Interview Question**

What is Amazon OpenSearch Service?

Answer:

Amazon OpenSearch Service is a managed service for OpenSearch (and legacy Elasticsearch). It is used for real-time search, log analytics, security analytics, and full-text search. It integrates with Kinesis Data Firehose, CloudWatch Logs, and Lambda for ingestion. OpenSearch Serverless automatically handles infrastructure scaling for variable workloads.

Q 266**Interview Question**

What is DynamoDB Provisioned vs On-Demand capacity?**Answer:**

Provisioned capacity mode requires you to specify read/write capacity units (RCUs/WCUs) in advance. Auto Scaling can adjust provisioned capacity within limits. On-Demand mode automatically scales to accommodate your traffic without planning. On-Demand is simpler and great for unpredictable workloads but more expensive at steady state. Provisioned is cost-effective for predictable workloads.

Q 267**Interview Question****What is RDS Snapshot and how to share it?****Answer:**

RDS manual snapshots are point-in-time backups that persist until manually deleted. To share a snapshot with another account, modify the snapshot's permissions to specify the account ID. To share across regions, copy the snapshot to the target region first. Encrypted snapshots require the KMS key to also be shared with the target account before the snapshot can be accessed.

Q 268**Interview Question****What is Amazon Aurora Global Database?****Answer:**

Aurora Global Database spans multiple AWS regions — one primary region handles writes, and up to 5 secondary regions receive read-only replicas with typically under 1 second replication lag. In a disaster scenario, you can promote a secondary region to primary (failover in under 1 minute). It is designed for globally distributed applications needing low-latency reads across continents.

Q 269**Interview Question****What is CloudWatch for RDS monitoring?****Answer:**

CloudWatch provides key RDS metrics: CPUUtilization, DatabaseConnections, FreeStorageSpace, ReadIOPS/WriteIOPS, ReadLatency/WriteLatency, FreeableMemory, and NetworkReceiveThroughput. Set alarms on low FreeStorageSpace and high CPUUtilization. Enable Enhanced Monitoring for OS-level details. Enable Performance Insights for query-level analysis and wait event visualization.

Q 270**Interview Question****What is RDS Performance Insights?****Answer:**

Performance Insights is a database performance tuning tool for RDS and Aurora. It visualizes database load as a graph of active sessions. You can filter by SQL query, host, user, and wait event to identify performance bottlenecks. The retention period is 7 days free and up to 2 years with the paid tier. It requires no agent installation — it works via the RDS API.

Section 10: Miscellaneous & Advanced Topics (Q271-Q300)**Q 271****Interview Question****What is AWS Well-Architected Framework?****Answer:**

The AWS Well-Architected Framework provides architectural best practices across five pillars: Operational Excellence (run and monitor systems), Security (protect data and systems), Reliability (recover from failures), Performance Efficiency (use resources efficiently), and Cost Optimization (avoid unnecessary costs). A sixth pillar, Sustainability, was added later. The Well-Architected Tool performs automated reviews.

Q 272**Interview Question****What is the AWS CAF (Cloud Adoption Framework)?****Answer:**

The AWS Cloud Adoption Framework provides guidance for organizations planning cloud migrations. It identifies perspectives across six areas: Business (strategy, portfolio), People (HR, change management), Governance (risk, compliance), Platform (infrastructure, CI/CD), Security (IAM, incident response), and Operations (monitoring, support). CAF helps identify skills gaps and plan migration journeys.

Q 273**Interview Question****What is AWS Landing Zone / Control Tower?****Answer:**

AWS Control Tower automates the setup of a secure, multi-account AWS environment (Landing Zone) following AWS best practices. It creates a master account, organizational structure, guardrails (mandatory and strongly recommended policies), a Service Catalog for account vending, and centralized logging and auditing — accelerating the creation of a compliant, governed multi-account environment.

Q 274**Interview Question****What is AWS Migration Hub?****Answer:**

AWS Migration Hub provides a central location to track the progress of application migrations across multiple AWS and partner migration tools. It integrates with Server Migration Service, Database Migration Service, and third-party tools. You can see the migration status of each application, even if different tools are used for different components.

Q 275**Interview Question**

What is AWS Application Discovery Service?

Answer:

Application Discovery Service helps plan migrations to AWS by collecting data about your on-premises servers — including configuration, performance, and network connections. The Agentless Discovery Connector (for VMware) collects data without installing agents. The Discovery Agent collects detailed OS-level data. This inventory data feeds into Migration Hub for planning.

Q 276**Interview Question**

What is AWS DMS (Database Migration Service)?

Answer:

AWS DMS helps migrate databases to AWS quickly and securely. The source database remains operational during migration, minimizing downtime. It supports homogeneous migrations (MySQL to RDS MySQL) and heterogeneous migrations (Oracle to Aurora PostgreSQL, using the Schema Conversion Tool). DMS supports continuous data replication for cut-over with minimal downtime.

Q 277**Interview Question**

What is AWS Server Migration Service (SMS)?

Answer:

AWS SMS automates, schedules, and tracks incremental replications of live server volumes to AWS. It converts on-premises VMware, Hyper-V, and Azure VMs to AMIs for launching on EC2. SMS creates incremental snapshots to keep the AMI up to date, minimizing the cut-over window. It is being replaced by AWS Application Migration Service (MGN).

Q 278**Interview Question**

What is AWS Application Migration Service (MGN)?**Answer:**

AWS MGN is the primary migration service for lift-and-shift migrations. It installs a lightweight replication agent on source servers that continuously replicates data to AWS staging. You launch test instances for validation, then perform a small-window cutover. MGN supports physical, virtual, and cloud-based sources — replacing SMS and the older CloudEndure Migration service.

Q 279**Interview Question****What is Multi-account strategy best practices?****Answer:**

Best practices include: separate accounts for production, staging, and development environments; separate accounts for security, logging, and networking; use AWS Organizations with SCPs; deploy infrastructure using CloudFormation StackSets; centralize logging in a dedicated log archive account; use AWS SSO (IAM Identity Center) for federated access; never use the root account for daily operations.

Q 280**Interview Question****What is AWS IAM Identity Center (SSO)?****Answer:**

AWS IAM Identity Center (formerly SSO) provides centralized access management for multiple AWS accounts and cloud applications. Users sign in once to the AWS access portal and access all permitted accounts and applications. It integrates with Active Directory, Azure AD, and other SAML 2.0 IdPs. For SysOps administrators managing multiple accounts, it significantly simplifies access management.

Q 281**Interview Question****What is AWS Snow Family?****Answer:**

The AWS Snow Family are physical devices for edge computing and data transfer. Snowcone (smallest, portable, 8 TB usable), Snowball Edge (Storage or Compute Optimized, up to 80 TB), and Snowmobile (a literal shipping container with up to 100 PB for massive migrations). Used when network transfer is impractical due to bandwidth limitations or data volume.

Q 282**Interview Question**

What is AWS Outposts?

Answer:

AWS Outposts brings native AWS infrastructure, services, APIs, and tools to virtually any on-premises facility. AWS delivers and installs Outposts hardware. You can run EC2, ECS, EKS, RDS, ElastiCache, and other services locally with the same AWS APIs — enabling hybrid architectures where data must remain on-premises for latency or data residency requirements.

Q 283**Interview Question**

What is Amazon Kinesis?

Answer:

Amazon Kinesis is a family of services for real-time data streaming. Kinesis Data Streams ingests real-time data (clickstreams, logs, IoT). Kinesis Data Firehose loads streaming data into S3, Redshift, OpenSearch, or Splunk automatically. Kinesis Data Analytics (now Managed Service for Apache Flink) enables real-time SQL or Apache Flink processing. Kinesis Video Streams handles video.

Q 284**Interview Question**

What is Amazon SQS?

Answer:

Amazon Simple Queue Service is a fully managed message queuing service that enables decoupling of distributed application components. Producers send messages to a queue; consumers poll and process them. SQS supports Standard queues (best effort ordering, at-least-once delivery) and FIFO queues (exactly-once processing, first-in-first-out). Message retention is 1-14 days.

Q 285**Interview Question**

What is Amazon SNS?

Answer:

Amazon Simple Notification Service is a managed pub/sub messaging service for application-to-application and application-to-person communication. Publishers send messages to topics, and SNS delivers to all subscribers (Lambda, SQS, HTTP/S endpoints, email, SMS, mobile push). SNS FIFO topics support ordered, deduplicated delivery. SNS is ideal for fan-out patterns — notify multiple systems on a single event.

Q 286**Interview Question**

What is AWS Local Zones?

Answer:

AWS Local Zones place AWS compute, storage, database, and networking services closer to large population centers that are far from AWS regions. They extend an AWS region to a metro area, providing single-digit millisecond latency for applications requiring very low latency access to end users — useful for media production, real-time gaming, and machine learning inference.

Q 287**Interview Question**

What is Amazon SES (Simple Email Service)?

Answer:

Amazon SES is a cloud-based email platform for sending transactional, marketing, and bulk emails at scale. SysOps administrators use SES for application-generated emails (password resets, notifications), CloudWatch alarm notifications, and billing alerts. SES provides email deliverability tools, bounce and complaint handling, and IP reputation management.

Q 288**Interview Question**

What is AWS Artifact?

Answer:

AWS Artifact is a self-service portal for on-demand access to AWS compliance documentation and AWS agreements. You can download SOC reports, PCI compliance documents, ISO certifications, and more. In AWS Artifact Agreements, you can review and accept agreements like HIPAA Business Associate Addendum (BAA) directly — important for regulated industries.

Q 289**Interview Question**

What is AWS Trusted Advisor cost optimization checks?

Answer:

Key cost optimization checks include: Low Utilization Amazon EC2 Instances (CPU <10% and network <5 MB), Idle Load Balancers (no active targets), Unassociated Elastic IP Addresses, Underutilized Amazon EBS Volumes, and Amazon RDS Idle DB Instances. These checks help identify resources that are incurring charges without delivering value — simple targets for cost reduction.

Q 290**Interview Question**

What is Disaster Recovery in AWS?

Answer:

AWS supports four DR strategies with increasing RTO/RPO performance: Backup & Restore (cheapest, hours RTO), Pilot Light (core services running, scale up on disaster, tens of minutes RTO), Warm Standby (scaled-down version always running, minutes RTO), and Multi-Site Active/Active (full-scale running in multiple regions, seconds RTO, most expensive).

Q 291**Interview Question**

What is Amazon EventBridge Scheduler?

Answer:

EventBridge Scheduler is a fully managed service for creating, running, and managing scheduled tasks at scale. Unlike CloudWatch Events cron rules, the Scheduler supports over 200 AWS services as targets, handles time zones natively, supports one-time and recurring schedules, and provides delivery guarantees with an event delivery window and retry policies.

Q 292**Interview Question**

What is AWS DataExchange?

Answer:

AWS Data Exchange makes it easy to find, subscribe to, and use third-party data in the cloud. Data providers publish data sets (financial data, geospatial data, weather data, demographic data) that customers can subscribe to and access directly in S3. It eliminates the need to build and maintain data transfer infrastructure between data providers and consumers.

Q 293**Interview Question**

What is AWS Well-Architected Tool?

Answer:

The AWS Well-Architected Tool is a free service in the console that helps you review the state of your workloads against the six pillars of the Well-Architected Framework. You answer a series of questions about your workload, and the tool identifies high-risk issues and provides improvement recommendations — serving as a structured architecture review process.

Q 294**Interview Question**

What is AWS Ground Station?

Answer:

AWS Ground Station is a fully managed service for controlling satellite communications, processing satellite data, and scaling operations without managing ground station infrastructure. You schedule contacts with satellites, and AWS Ground Station downlinks telemetry to your VPC. It enables businesses in earth observation, weather forecasting, and communications to access satellite data without building their own ground station network.

Q 295**Interview Question**

What is Amazon AppFlow?

Answer:

Amazon AppFlow is a fully managed integration service for securely transferring data between AWS services and SaaS applications like Salesforce, SAP, Slack, ServiceNow, and Marketo. You configure flows specifying source, destination, transformation rules, and schedule. AppFlow encrypts data in transit and at rest, ensuring compliance when moving business data.

Q 296**Interview Question**

What is AWS Resilience Hub?

Answer:

AWS Resilience Hub provides a central place to define, validate, and track resilience of your applications. You import application resources, set Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets, and the hub assesses your application against those targets. It identifies gaps and provides specific recommendations to improve resilience against disruptions.

Q 297**Interview Question**

What is Amazon Macie classification job?

Answer:

A Macie classification job scans specified S3 buckets for sensitive data. You define the scope (specific buckets, prefixes, or object criteria), schedule (one-time or periodic), and sampling depth. Macie uses ML and pattern matching to identify PII, financial data, and credentials. Findings are visible in the Macie console, Security Hub, and can be sent to EventBridge for automated response.

Q 298**Interview Question**

What is AWS Config Conformance Packs?

Answer:

Conformance Packs are collections of AWS Config rules and remediation actions packaged together as a single entity. AWS provides pre-built conformance packs for standards like CIS, PCI DSS, HIPAA, and NIST. You deploy them to an account or organization using CloudFormation-style templates — enabling bulk deployment of compliance frameworks across your entire AWS environment.

Q 299**Interview Question**

What is Amazon Forecast?

Answer:

Amazon Forecast is a fully managed ML service for generating accurate time series forecasts. It uses the same technology AWS uses to forecast Amazon.com demand. For SysOps scenarios, it can forecast resource demand for proactive capacity planning, predict cost trajectories, or anticipate traffic patterns — enabling smarter capacity and cost management decisions.

Q 300**Interview Question**

What are the key AWS SysOps Administrator exam domains?

Answer:

The AWS SysOps Administrator Associate exam (SOA-C02) covers: Monitoring, Logging, and Remediation (20%), Reliability and Business Continuity (16%), Deployment, Provisioning, and Automation (18%), Security and Compliance (16%), Networking and Content Delivery (18%), and Cost and Performance Optimization (12%). Hands-on labs (exam labs) are included in the new exam format.